POLITECNICO DI TORINO

Bachelor's Degree in Management Engineering



Bachelor's Thesis in Statistics

Hidden Markov Models for Capturing Uncertainty in Financial Markets

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Sommario

Le dinamiche dei mercati finanziari sono caratterizzate da una complessa instabilità temporale: la dispersione dei rendimenti cambia nel tempo, la volatilità tende a concentrarsi in fasi persistenti e gli eventi estremi si verificano più frequentemente di quanto suggerirebbe un modello gaussiano statico. Questa tesi si interroga su *fino a che punto* la statistica possa descrivere tale incertezza, distinguendo le regolarità modellabili dalla componente casuale che inevitabilmente permane nei mercati reali.

Il quadro teorico formalizza l'informazione in forma matematica e distingue i dati *osservabili* dalle condizioni di mercato *latenti*. Vengono introdotte le catene di Markov per modellare le transizioni tra stati e successivamente gli Hidden Markov Models (HMM), nei quali il regime di mercato non è direttamente osservabile. L'analisi utilizza quindi specifici algoritmi di inferenza: Forward-Backward per stimare le probabilità degli stati, Viterbi per ricostruire la sequenza di regimi più probabile e Baum-Welch per stimare i parametri a partire dai rendimenti osservati.

L'analisi empirica applica in Python un HMM gaussiano a tre stati ai rendimenti logaritmici giornalieri dell'S&P 500 dal 1980 al 2024. Il modello stima le probabilità di permanenza e transizione tra regimi di bassa, media e alta volatilità, analizzando la distribuzione dei rendimenti in ogni stato, confrontandola con i pattern introdotti.

I risultati empirici sono coerenti con l'impostazione adottata: i regimi stimati presentano livelli di volatilità distinti e una persistenza compatibile con il *volatility clustering*, mentre la loro combinazione offre un confronto più flessibile rispetto a un modello gaussiano statico. L'HMM non elimina la casualità intrinseca dei mercati; ne offre una quantificazione condizionata al modello stimato attraverso probabilità di stato e dinamiche di regime, pur rimanendo un'approssimazione statistica.

Abstract

Financial markets exhibit pronounced changes in their behavior over time: return distributions evolve, volatility tends to persist in clusters and extreme events occur more often than a static Gaussian model would suggest. This thesis asks *how far* statistics can go in describing that uncertainty, separating patterns that can be modeled from the random component that inevitably remains in actual markets.

The theoretical framework explains how information flows over time and distinguishes *observable* data from *hidden* market conditions. It introduces Markov chains to model transitions between states and extends them to Hidden Markov Models (HMMs), where the regime is unobserved. The analysis uses specific inference algorithms: Forward-Backward to estimate state probabilities, Viterbi to reconstruct the likely regime path and Baum-Welch to estimate parameters from observed returns.

The empirical analysis implements a three-state Gaussian HMM in Python on daily S&P 500 log-returns from 1980 to 2024. The model estimates the probabilities of remaining in or moving between low, medium and high-volatility regimes, describing the return distribution within each state, comparing it with the introduced patterns.

The empirical findings are consistent with the analytical approach developed in this thesis: the fitted regimes exhibit distinct volatility levels and persistence compatible with *volatility clustering*, while their mixture provides a descriptive comparison with a static Gaussian benchmark. The HMM does not remove inherent market randomness; rather, it provides a model-based quantification of uncertainty through state probabilities and regime dynamics, while remaining a statistical approximation.

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Section I

Financial Framework

Chapter 1

Market Dynamics in a Non-Stationary World

1.1 Financial Time Series, Prices and Returns

Financial markets are complex, dynamic systems where prices continuously evolve under the influence of economic forces, investor behavior, and unexpected shocks. Understanding this evolution requires moving beyond isolated observations and embracing a structured and time-indexed framework. The foundation of any rigorous quantitative analysis is, therefore, not a single data point but a financial time series[1].

Definition 1.1 (Financial Time Series). A financial time series is a sequence of numerical observations recorded at successive points in time, denoted as:

$$\{x_t\}_{t=0}^{T} \tag{1.1}$$

where x_t represents the observed value at time t and T is the final observation.

The quantity x_t can represent asset prices, returns, volatility estimates, trading volumes or any other market variable. In this thesis, we will work primarily with the price series $\{S_t\}_{t=0}^{T}$ and the derived logarithmic return series $\{r_t\}_{t=1}^{T}$.

Prices reflect the cumulative effect of all market movements, while returns isolate the incremental changes between consecutive observations. For this reason, the analysis is usually carried out on logarithmic returns rather than on raw prices[2].

Definition 1.2 (Simple and Logarithmic Return). Given two consecutive prices S_{t-1} and S_t, the simple return R_t and the logarithmic return r_t are defined respectively:

$$R_t = \frac{S_t - S_{t-1}}{S_{t-1}}, \qquad r_t = \ln\left(\frac{S_t}{S_{t-1}}\right). \tag{1.2}$$

Their additivity across time horizons linearizes cumulative asset growth, facilitating long-term evaluation. Let $r_{0,k}$ be the cumulative log-return from time 0 to time k:

$$r_{0,k} = \sum_{t=1}^{k} r_t = \ln(S_k) - \ln(S_0) \tag{1.3}$$

Figures 1.1 and 1.2 illustrate the difference between the raw price series of the S&P 500 index and its daily logarithmic returns. While raw equity indices wander unpredictably over an irregular trend, the logarithmic transformation compresses these movements into dimensionless changes that fluctuate around a near zero mean.

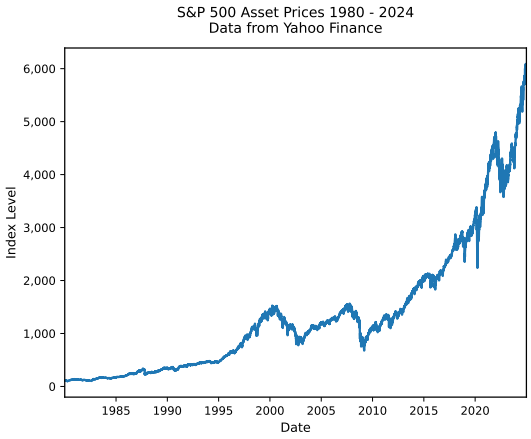


Figure 1.1: *Raw Price Series*

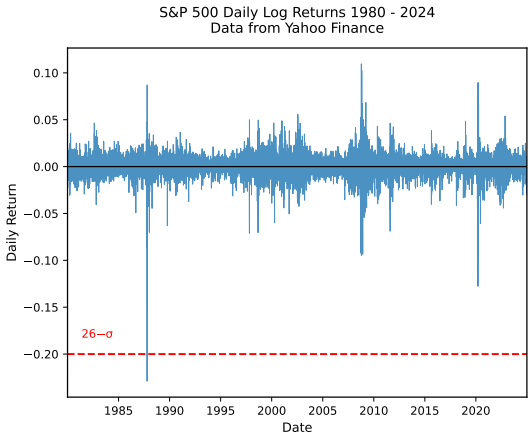


Figure 1.2: *Logarithmic Returns*

The raw price series exhibits an upward, irregular trend, limiting comparability across periods. The logarithmic transformation stabilizes the mean and highlights variance shifts, notably the most extreme downward spike from the 1987 Black Monday, a 26σ event.

The graphs clearly show that, although the logarithmic transformation stabilizes the sample mean $\hat{\mu}$, the sample variance $\hat{\sigma}^2$ is not constant. The changing amplitude of the oscillations reflects **time-varying conditional dynamics**, challenging the constant variance assumptions often used in standard classical financial models.

To formalize this, we introduce the concept of a stochastic process: a collection of random variables indexed by time, $\{X_t\}_{t\in T}$, where each X_t is defined on a common probability space. To evaluate its stability, we use the concept of stationarity[1].

Definition 1.3 (Stationarity). A stochastic process $\{X_t\}$ is **weakly stationary** if its mean is constant over time ($\mathbb{E}[X_t] = \mu$), its variance is finite and constant ($\mathrm{Var}(X_t) = \sigma^2 < \infty$) and its autocovariance depends exclusively on the lag τ ($\mathrm{Cov}(X_t, X_{t+\tau}) = \gamma(\tau)$). Additionally, a process is **strongly stationary** if its entire joint probability distribution remains unchanged under any shift in time.

Visually, if the mean represents the central baseline and the variance dictates the vertical amplitude, the autocovariance captures the linear memory of the series. Requiring it to depend only on the lag means the statistical rhythm between consecutive observations remains identical regardless of the observed time period.

The S&P 500 return series displays pronounced changes in dispersion: while the sample mean remains approximately stable, periods of calm alternate with periods of stress. This empirical pattern challenges a simple constant-variance representation and suggests time-varying conditional dynamics, motivating the development of models able to accommodate changing volatility across market regimes.

To capture the distributional effects of time-varying volatility, relying solely on the first two moments, mean and variance, is insufficient. A complete characterization requires the full spectrum of sample moments, including the higher-order ones[3].

Definition 1.4 (Sample Moments of Returns). Given a sample of logarithmic returns $\{r_t\}_{t=1}^{T}$, the measures of central tendency and dispersion are defined as:

$$\hat{\mu} = \frac{1}{T}\sum_{t=1}^{T} r_t, \qquad \hat{\sigma}^2 = \frac{1}{T-1}\sum_{t=1}^{T}(r_t - \hat{\mu})^2. \tag{1.4}$$

Aggregating these returns into a single empirical distribution inevitably distorts its shape, necessitating measures of the resulting asymmetry and tail weight:

$$\widehat{\mathrm{skew}} = \frac{1}{T}\sum_{t=1}^{T}\left(\frac{r_t - \hat{\mu}}{\hat{\sigma}}\right)^3, \qquad \widehat{\mathrm{kurt}_{ex}} = \frac{1}{T}\sum_{t=1}^{T}\left(\frac{r_t - \hat{\mu}}{\hat{\sigma}}\right)^4 - 3. \tag{1.5}$$

The sample variance can be adapted into a dynamic **rolling variance**, expression of the time-varying behavior. Evaluated over a window of n days, it is defined as:

$$\hat{\sigma}_t^2 = \frac{1}{n-1} \sum_{i=0}^{n-1} (r_{t-i} - \hat{\mu}_t)^2 \tag{1.6}$$

where $\hat{\mu}_t$ is the rolling sample mean computed over the specific window n.

The square root of this variance is **volatility**, the standard deviation of returns and a primary measure of market risk, capturing the magnitude of price movements in both directions. Its analysis is central to modern finance, governing asset pricing, portfolio selection and risk management through a fundamental principle: elevated volatility demands greater compensation from investors holding unstable assets.

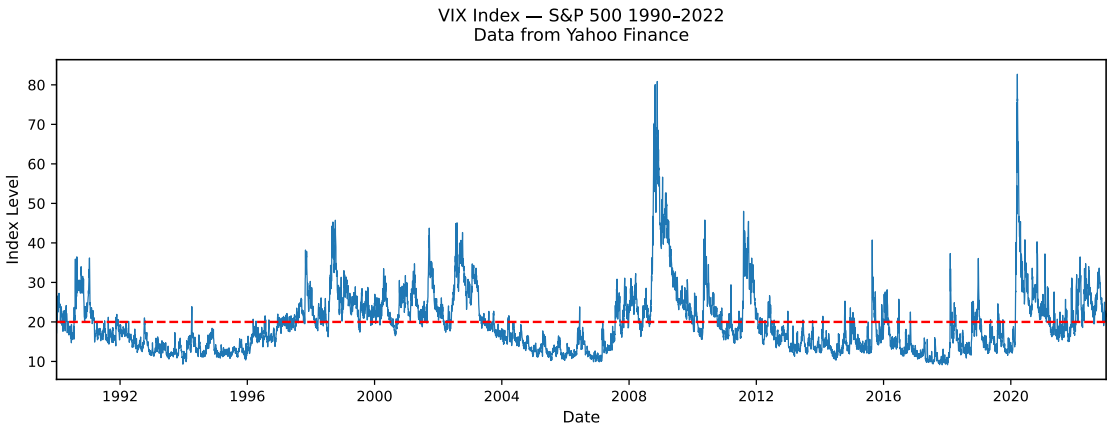


Figure 1.3: *Path of the Volatility Index (VIX), a leading measure of market expectations of near-term volatility. The gap between the reference level (red dashed line) and the observed values illustrates the limits of a constant-volatility representation.*

Accurately assessing this risk is challenged by the fact that volatility is not constant: it fluctuates across time, as a sign of time-varying market conditions. Traditional models often assume a fixed variance, which is contradicted by the empirical record: the 1987 Black Monday crash is represented as a 26-sigma event under a static Gaussian benchmark, illustrating the near-zero probability assigned to such shocks.

These empirical irregularities motivate the implementation of models able to account for changing distributional conditions across distinct *volatility regimes*.

1.2 Volatility Clustering and Leptokurtosis

The dynamic nature of market variance, established in the previous section, directly governs the shape of financial returns. When empirical financial time series are examined, these time-varying dynamics are reflected in two deeply connected distributional phenomena, namely **volatility clustering** and **leptokurtosis**.

The phenomenon of volatility clustering refers to the temporal persistence of market turbulence[6, 7]. As observed by the mathematician Benoît Mandelbrot, pioneer in the study of financial time series, financial markets exhibit a remarkable regularity:

> *"Large changes tend to be followed by large changes, of either sign, and small changes tend to be followed by small changes."*[4]

This observation suggests that market shocks are not isolated events; instead, periods of high or low activity persist locally before eventually transitioning. To rigorously measure this persistence, quantitative analysis relies on the sample autocorrelation function (ACF), which evaluates this dependence between successive observations[1].

Definition 1.5 (Sample Autocorrelation Function). Given a generic financial time series $\{x_t\}_{t=1}^{T}$, the sample autocorrelation function (ACF) at lag $\tau \geq 1$ measures the linear dependence between the series and a time-shifted version of itself, defined as:

$$\hat{\rho}(\tau) = \frac{\sum_{t=1}^{T-\tau}(x_t - \hat{\mu})(x_{t+\tau} - \hat{\mu})}{\sum_{t=1}^{T}(x_t - \hat{\mu})^2} \tag{1.7}$$

where τ denotes the lag, $\hat{\mu}$ is the sample mean of the series, and $\hat{\rho}(\tau) \in [-1, 1]$.

Applying this metric to logarithmic returns $\{r_t\}$, we typically observe that the sample autocorrelation at any lag $\tau \geq 1$ is statistically close to zero:

$$\hat{\rho}_r(\tau) \approx 0 \tag{1.8}$$

This lack of serial correlation indicates an absence of linear memory: knowing the market's direction today provides limited linear information for predicting tomorrow.

This observation does not establish independence or the **Markov property**, which will be introduced in Chapter 3. Rather, it separates the limited linear predictability of return direction from the stronger persistence observed in return magnitude.

From a statistical perspective, if market movements were truly independent and random events, taking their absolute value would still result in an autocorrelation of zero. However, empirical financial data reveals a completely different reality:

$$\hat{\rho}_{|r|}(\tau) > 0 \tag{1.9}$$

Unlike raw returns, absolute returns exhibit a positive and **slowly decaying autocorrelation**. This statistical divergence represents the statistical evidence of volatility clustering: while the *direction* of returns is unpredictable, the *magnitude* of those price changes exhibits strong historical persistence.

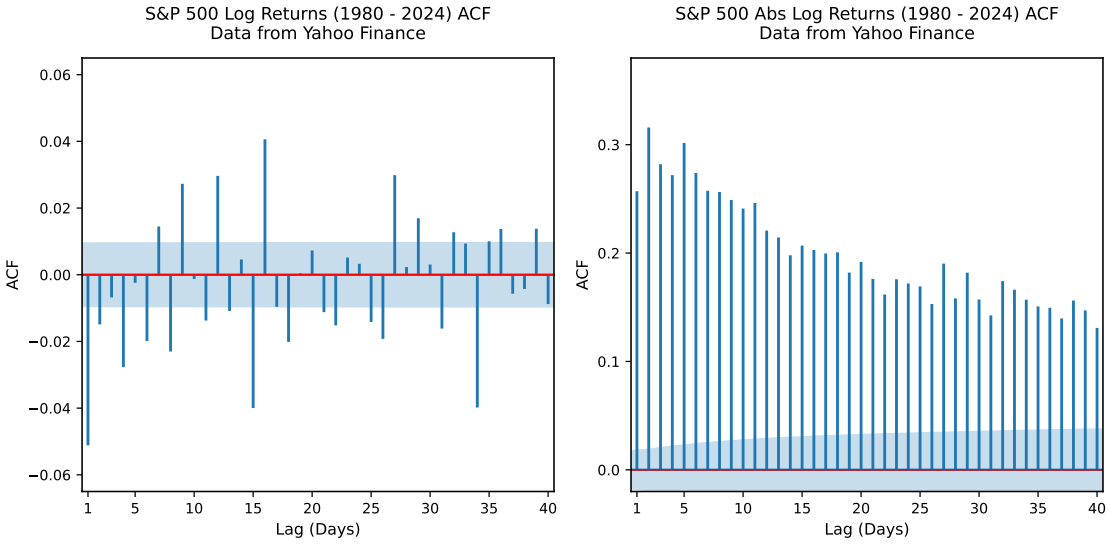


Figure 1.4: *The left panel shows autocorrelation values close to zero, confirming the absence of linear dependence in raw returns. The right panel shows consistently positive values for absolute returns, revealing a strong persistence in magnitude.*

Leptokurtosis: The Distributional Consequence

A related distributional consequence of volatility clustering is the phenomenon of leptokurtosis, commonly known as *the fat tails problem*. When a dynamic, time-varying return series is aggregated into a single probability histogram, it creates an unconditional distribution that systematically deviates from a Gaussian bell curve.

The resulting unconditional distribution exhibits a higher and narrower peak around the mean, due to frequent small returns during stable periods, and heavier tails, from the clustered shocks of turbulent phases. By plotting them all together, distinct volatility environments become artificially compressed into one static snapshot.

As established in Equation 1.5, which we recall, this distributional feature is formally captured by the fourth standardized moment, namely the **excess kurtosis**:

$$\widehat{\text{kurt}}_{ex} = \frac{1}{T}\sum_{t=1}^{T}\left(\frac{r_t - \hat{\mu}}{\hat{\sigma}}\right)^4 - 3 \tag{1.5}$$

For a theoretical Gaussian distribution, $\widehat{\text{kurt}}_{ex} = 0$ by construction. In contrast, empirical financial returns consistently exhibit $\widehat{\text{kurt}}_{ex} \gg 0$, mathematically confirming that the probability mass in the tails significantly exceeds the Gaussian benchmark.

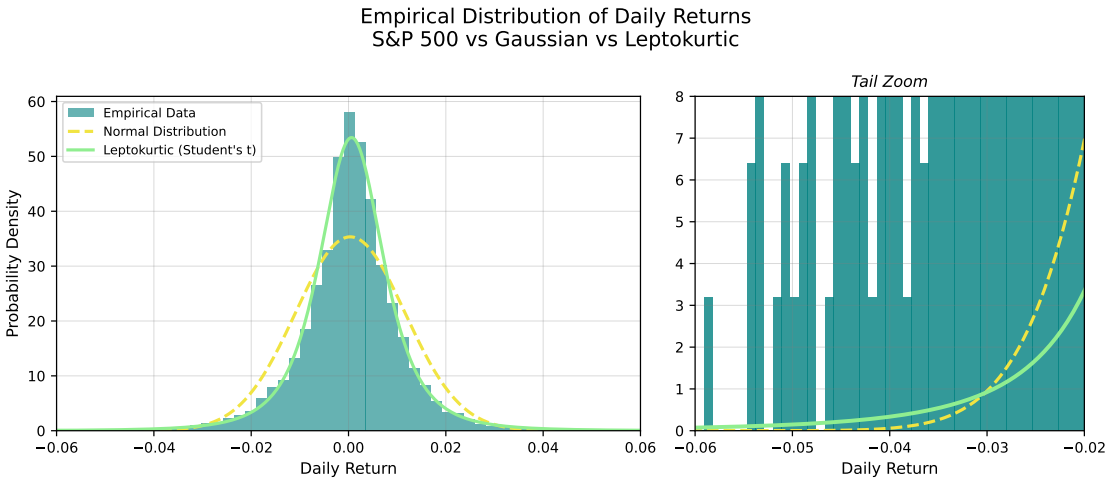


Figure 1.5: *The comparison between empirical log returns, the Normal distribution and the fitted Student's t distribution ($\nu = 3$ degrees of freedom, estimated from data using Python) shows how the Gaussian curve assigns insufficient probability mass to tails, thereby underestimating the likelihood of extreme events like the 1987 crash.*

To understand how this leptokurtic shape emerges from aggregating distinct market environments, the distribution of returns can be modeled as a **finite Gaussian mixture**[5], a flexible framework that accommodates both the peak and the tails.

Definition 1.6 (Finite Gaussian Mixture). A return variable R follows a finite Gaussian mixture with J components if its density can be written as:

$$f_R(r) = \sum_{j=1}^{J} \pi_j \varphi(r; \mu_j, \sigma_j^2) \tag{1.10}$$

where $\pi_j \geq 0$ are the mixing weights such that $\sum_{j=1}^{J} \pi_j = 1$, and $\varphi(r; \mu_j, \sigma_j^2)$ denotes the Gaussian density with mean μ_j and variance σ_j^2.

If one component captures a calm regime with low variance and another captures a stressed regime with large dispersion, their aggregate sum inherently displays a non-Gaussian distribution, with a sharp central peak and heavy tails, even though each of the underlying components is individually normally distributed.

While the Gaussian mixture successfully reconstructs the leptokurtic shape, it remains a purely *static* framework. The mixing weights π_j are fixed and time-independent, implying that the model assumes the market randomly jumps between states at each time step, ignoring the temporal persistence of volatility clustering.

Financial markets present a **dual mathematical challenge**: their unconditional distribution is strictly non-Gaussian and their conditional behavior changes over time. This limitation of classical assumptions motivates the necessity of a dynamic approach that models the latent states that actually govern market transitions.

1.3 Limitations of Gaussian Models

The dual challenge of non-Gaussianity and non-stationarity fundamentally weakens the traditional framework of financial analysis. This classical paradigm is deeply rooted in the assumption of absolute stability, relying on two foundational pillars of statistical inference: the Law of Large Numbers and the Central Limit Theorem.

The **Law of Large Numbers** states that the sample average of a sequence of independent and identically distributed random variables, $\{X_1, X_2, \ldots, X_n\}$ with $E[X_i] = \mu$, converges to the expected value as the number of observations increases.

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow{p} \mu \quad \text{as } n \to \infty \tag{1.11}$$

The **Central Limit Theorem** dictates that the sample mean of a large number of independent and identically distributed random variables, each with finite variance σ^2, converges to a normal distribution regardless of the distribution of the variables.

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} \mathcal{N}(0, \sigma^2) \quad \text{as } n \to \infty \tag{1.12}$$

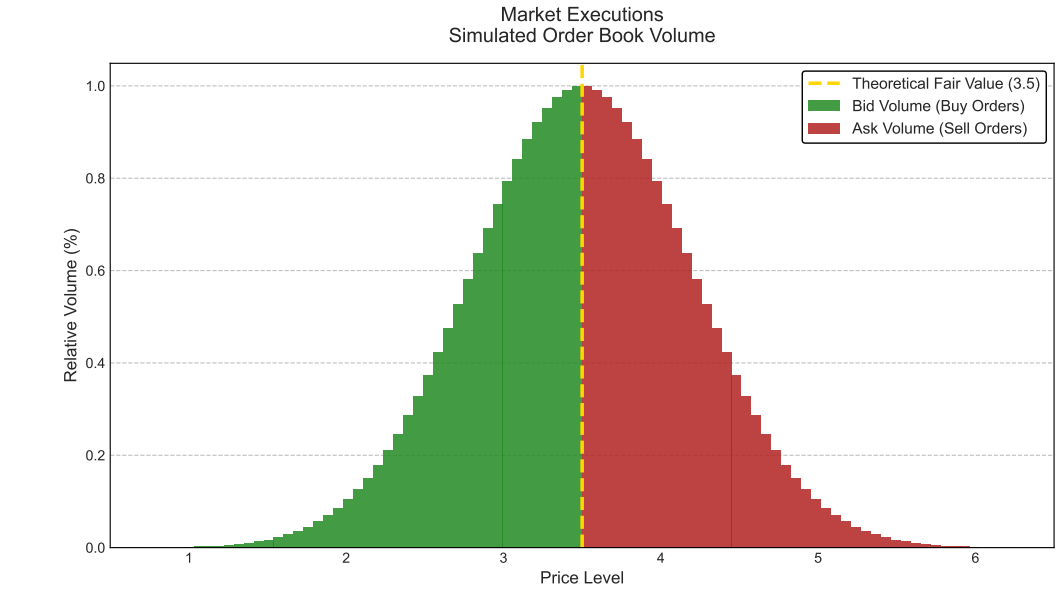
The application of the *i.i.d.* versions of these results is not automatically justified for financial returns, which may exhibit dependence and time-varying distributions. Volatility clustering challenges the independence assumption, while time-varying distributional features could violate the *identical-distribution* assumption.

To make the severity of this failure intuitive, consider the analogy of a fair six-sided die,* with an equal probability $P(X = x) = 1/6$ for $x \in \{1, \ldots, 6\}$ and a zero probability mass everywhere else. In a perfectly stable environment, the outcome of any single roll is random, but the theoretical expected value is strictly constant:

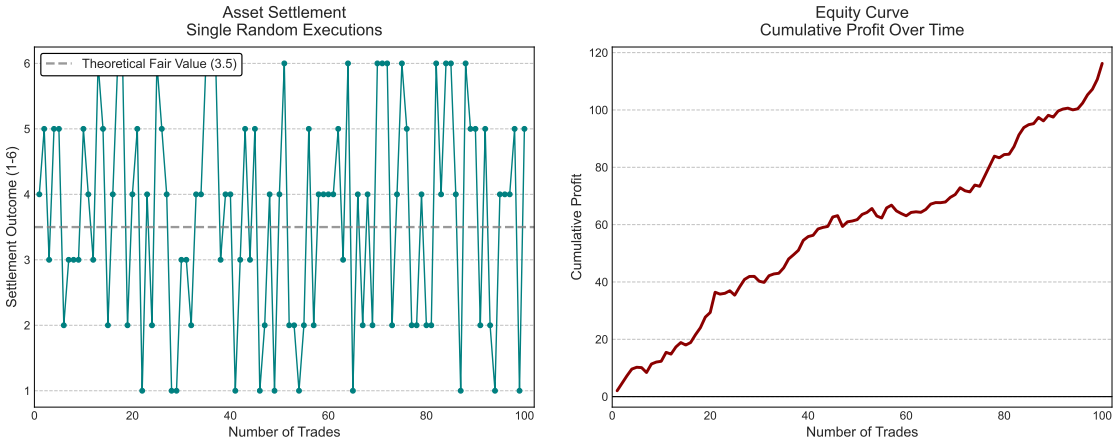
$$\mathbb{E}[X] = \sum_{i=1}^{6} i\, P(i) = 3.5 \tag{1.13}$$

At each trade, the die determines the random final asset value $Y_t \in \{1, \ldots, 6\}$. A market-making strategy places the bid b_t below fair value 3.5 and the ask price a_t above it. A trade generates profit $\Pi_t = Y_t - b_t$ when buying at bid price b_t, and $\Pi_t = a_t - Y_t$ when selling at ask price a_t. The equity curve is cumulative profit, $W_n = W_0 + \sum_{t=1}^{n} \Pi_t$. Since $b_t < 3.5 < a_t$, expected average profit is positive; under fixed probabilities, the LLN supports an upward trend despite drawdowns.

*Illustrative setup adapted from Roman Paolucci, *Non-Stationarity and Why Market Timing Fails*, 2026; see the *Quant-Guild-Library* repository.



(a)



(b) **(c)**

Figure 1.6: *Simulation of a market maker in a strictly stationary environment.* ***(a)*** *Knowing the theoretical fair value (3.5), the market maker places bid volumes below this level and ask volumes above it, concentrating it around the center of the distribution.* ***(b)*** *The settlement value of each trade, simulated as a die roll, is random and uniformly distributed, illustrating short-term unpredictability.* ***(c)*** *The equity curve provides an illustration of cumulative profit under a positive expected edge. Thus, the LLN explains why average outcomes stabilize over repeated trials.*

However, regime changes in financial markets can be interpreted as the die changing while the game is being played. If the market abruptly replaces the six-sided die with a ten-sided one, a trader relying on parameters from the previous regime will likely experience catastrophic losses, as historical data no longer reflect current risks.

To survive this environment, we must adopt **regime-switching models**[11], frameworks designed to identify which die the market is currently rolling. By allowing the underlying parameters to shift dynamically, these models address the dual challenge of non-Gaussianity and non-stationarity established through this chapter.

Because these distinct market states are strictly latent and information unfolds sequentially over time, we must first build a rigorous framework to handle time-evolving knowledge. This logical necessity dictates the transition to Chapter 2, where we introduce the measure-theoretic concepts of filtrations and latent variables.*

*The theoretical background and definitions follow Mastrantonio [1], Semeraro [2] and Chiang [3]. Additional references include Cont [7], Zanon [6], Mandelbrot [4], Oosterlee and Grzelak [5] and Hamilton [11]. Numerical superscripts refer to the complete Bibliography placed at the end.

Chapter 2

Information, Filtration and Latent Variables

2.1 Filtration $\{\mathcal{F}_t\}$ and Temporal Information

A rigorous modeling of the continuous evolution of financial markets requires a mathematical framework that formalizes the sequential flow of time and the progressive accumulation of information. In stochastic analysis, this role is fulfilled by the measure-theoretic concepts of **probability spaces** and **filtrations**, built upon the triplet known as the probability space $(\Omega, \mathcal{F}, \mathbb{P})$[8].

Definition 2.1 (Probability Space). A probability space consists of three elements:

1. Ω*, the sample space,* representing the set of all possible outcomes of the system, namely all possible future paths of a financial asset.

2. $\mathcal{F}$*, a* σ*-algebra on* Ω*,* which is a collection of subsets of Ω containing $\emptyset$ and closed under complements and countable unions.

3. $\mathbb{P}$*, a probability measure,* which assigns to each $A \in \mathcal{F}$ a value [0,1], satisfies $\mathbb{P}(\Omega) = 1$, and is countably additive on disjoint events.

In a financial context, the σ-algebra $\mathcal{F}$ represents the full set of possible events, while at time t only a subset of accumulated knowledge, generated by past prices, disclosed returns and volumes, is available. This progressive flow is mathematically formalized as a *filtration*, which encapsulates all that is known up to time t[9].

Definition 2.2 (Filtration). Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and T an index set. A *filtration* $\mathbb{F} = \{\mathcal{F}_t\}_{t \in T}$ is a collection of sub-σ-algebras of $\mathcal{F}$ such that:

$$\mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F}, \quad \text{for all } s \leq t \tag{2.1}$$

The initial sub-σ-algebra $\mathcal{F}_0$ represents knowledge before any sequential observations. For past times $s \leq t$, $\mathcal{F}_s$ captures information accumulated up to that point, and $\mathcal{F}_t$ is total information at t. The nesting property $\mathcal{F}_s \subseteq \mathcal{F}_t$ ensures information accumulates over time, revealing the market's true state without losing past data.

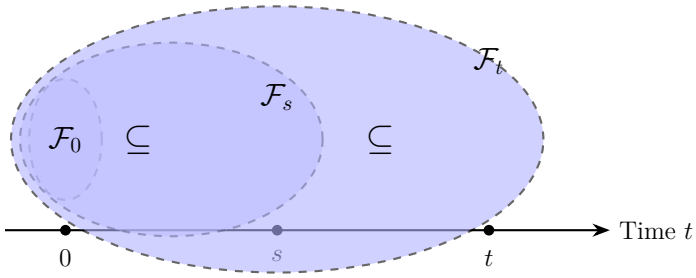


Figure 2.1: *Representation of a filtration $\mathbb{F} = \{\mathcal{F}_t\}_{t \geq 0}$. As time progresses along the horizontal axis, the σ-algebras expand ($\mathcal{F}_0 \subseteq \mathcal{F}_s \subseteq \mathcal{F}_t$), illustrating the accumulation of historical data and the set of events with known outcomes to the market observer.*

To ensure the continuous flow of time, market variables such as prices and returns cannot be treated as static elements. Instead, they must evolve in step with the information available at each moment, described by the concept of *adapted process*[10].

Definition 2.3 (Adapted Process). Recalling the definition of a stochastic process from Section 1.1, we say that a stochastic process $X = \{X_t\}_{t \in T}$ is *adapted* to the filtration $\{\mathcal{F}_t\}_{t \in T}$ if, for every $t \in T$, the random variable X_t is $\mathcal{F}_t$-measurable.

Let us define a financial process of interest, $X = \{X_t\}_{t \geq 0}$, representing the logarithmic returns of a given asset. The fundamental information structure associated with this asset is the *natural filtration* generated by the process itself, defined as:

$$\mathcal{F}_t^X = \sigma(X_s : 0 \leq s \leq t) \tag{2.2}$$

By definition of the generated σ-algebra $\sigma(\cdot)$, $\mathcal{F}_t^X$ is the smallest σ-algebra to which X_t is adapted, containing data revealed by the asset's historical trajectory. Given the available information, a mathematical operator is required for predictions[9].

Definition 2.4 (Conditional Expectation). Let Y be an integrable random variable, $\mathbb{E}[|Y|] < \infty$, defined on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$, and let $\mathcal{F}_t \subseteq \mathcal{F}$ be a sub-σ-algebra. The *conditional expectation* of Y given $\mathcal{F}_t$, denoted by $\mathbb{E}[Y \mid \mathcal{F}_t]$, is the unique $\mathcal{F}_t$-measurable variable Z that provides the best unbiased estimate:

$$\int_A Z \, d\mathbb{P} = \int_A Y \, d\mathbb{P}, \quad \text{for all } A \in \mathcal{F}_t. \tag{2.3}$$

This operator formalizes the optimal forecast of a future random variable Y, isolating its predictable component from uncertainty generated by new information $\notin \mathcal{F}_t$.

When the variable Y has finite variance, $Y \in L^2(\Omega)$, the space of random variables acquires a geometric structure. In this framework, the conditional expectation $\mathbb{E}[Y \mid \mathcal{F}_t]$ can be interpreted as the orthogonal projection of Y onto the subspace of $\mathcal{F}_t$-measurable random variables, minimizing $\mathbb{E}[(Y - Z)^2]$ over all $\mathcal{F}_t$-measurable Z.

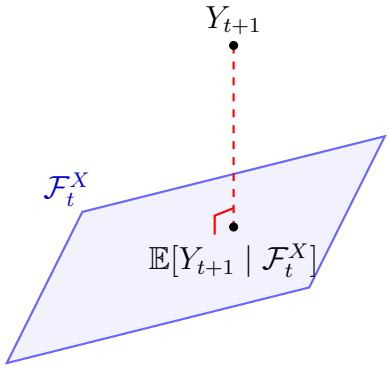


Figure 2.2: *Conditional expectation as an orthogonal projection. The filtration $\mathcal{F}_t^X$ defines a restricted information plane, while the conditional expectation is the projection of Y_{t+1} onto available information. Since Y_{t+1} may also depend on latent factors outside that plane, the residual reflects the limits of observable data.*

In a perfectly stationary world, the true future state Y_{t+1} would lie very close to our observable plane, making the forecasting error negligible. However, since financial markets are continually subjected to structural changes, attempting to forecast Y_{t+1} using only the natural filtration of prices, $\mathcal{F}_t^X$, fails to capture these latent dynamics.

To illustrate these principles, we apply the conditional expectation to two problems: the *direction of prices* and the *magnitude of risk*. How estimates evolve over time is captured by the Tower Property. For any $s \leq t$ (so that $\mathcal{F}_s \subseteq \mathcal{F}_t$) and integrable Y:

$$\mathbb{E}\Big[\mathbb{E}[Y \mid \mathcal{F}_t] \mid \mathcal{F}_s\Big] = \mathbb{E}[Y \mid \mathcal{F}_s]. \tag{2.4}$$

Definition 2.5 (Martingale)[9]. A stochastic process $M = \{M_t\}_{t\geq 0}$ is a martingale with respect to a filtration $\{\mathcal{F}_t\}_{t\geq 0}$ and a probability measure $\mathbb{P}$ if it satisfies:

1. **Adaptedness:** M_t is $\mathcal{F}_t$-measurable and the current value is fully known;

2. **Integrability:** $\mathbb{E}[|M_t|] < \infty$ for all t, meaning the expected value exists;

3. **Fair Game:** $\mathbb{E}[M_{t+1} \mid \mathcal{F}_t] = M_t$ almost surely, establishing that the optimal unbiased prediction of the future state is simply the current state itself.

The probability measure $\mathbb{P}$ governs the dynamics of asset prices, returns and volatility, allowing for empirical modeling of observed market behavior and time-varying risk. The risk-neutral measure $\mathbb{Q}$ is used to price financial derivatives avoiding arbitrage[10]. In this thesis, all subsequent analysis is conducted under the real-world measure $\mathbb{P}$.

While asset prices may exhibit non-zero expected returns over longer horizons, linear predictability at short horizons is rather limited. Relative to the observable filtration $\mathcal{F}_t^X$, past prices may therefore provide limited predictive information about the next return. In this sense, short-horizon returns may exhibit a martingale-like lack of linear predictability, motivating the conditional approximation:

$$\mathbb{E}[S_{t+1} \mid \mathcal{F}_t^X] \approx S_t, \quad \mathbb{E}[r_{t+1} \mid \mathcal{F}_t^X] \approx 0 \tag{2.5}$$

Having introduced this martingale intuition for short-horizon returns, the focus shifts to the second conditional moment, variance. Unlike price direction, volatility exhibits strong temporal persistence, formally defined as:

$$Var(r_{t+1} \mid \mathcal{F}_t^X) = \mathbb{E}\left[(r_{t+1} - \mathbb{E}[r_{t+1} \mid \mathcal{F}_t^X])^2 \mid \mathcal{F}_t^X\right] \tag{2.6}$$

This conditional variance highlights the structural limitations of the observable filtration $\mathcal{F}_t^X$. As suggested by the geometric analogy, market turbulence cannot be fully explained by historical information alone. Therefore, financial time series analysis is not a problem of perfect foresight, but rather of **incomplete information**.

Because observed price history is insufficient to capture the underlying market dynamics, the framework must account for a formal separation between observable market data and unobserved latent factors that must be inferred probabilistically.

2.2 Observable and Latent Variables

The discrepancy between the unpredictable directional movements of financial returns and the persistent nature of their magnitude necessitates a mathematical formalization. To model a market where fundamental risk drivers are not directly observed, we partition the filtration $\{\mathcal{F}_t\}_{t\geq 0}$ into observed and hidden components, whose combined information is represented via the join operator $\vee$[8].

Definition 2.6 (Join of σ-Algebras). The join of two sub-σ-algebras $\mathcal{G}_1$ and $\mathcal{G}_2$ on a common probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is the smallest σ-algebra containing both:

$$\mathcal{G}_1 \vee \mathcal{G}_2 = \sigma(\mathcal{G}_1 \cup \mathcal{G}_2) \tag{2.7}$$

Given the complete information set $\mathcal{F}_t$ at time t, its fundamental partition is:

$$\mathcal{F}_t = \mathcal{F}_t^{\text{obs}} \vee \mathcal{F}_t^{\text{lat}} \tag{2.8}$$

In this formulation, $\mathcal{F}_t^{\text{obs}}$ represents the sub-σ-algebra generated by all the data directly recorded by the market. If the only observable data is the historical return sequence X, then $\mathcal{F}_t^{\text{obs}} \equiv \mathcal{F}_t^X$. Conversely, $\mathcal{F}_t^{\text{lat}}$ represents the sub-σ-algebra generated by the underlying forces that influence market behavior. This decomposition allows us to formally classify the random variables that populate financial models[11].

Definition 2.7 (Observable Variable). A random variable Y_t is said to be observable at time t if it is $\mathcal{F}_t^{\text{obs}}$-measurable, meaning that its realized value $Y_t(\omega)$ is fully determined by the information available to the market at time t. Typical examples in financial time series include logarithmic returns r_t, prices S_t and trading volumes.

However, the assumption that market dynamics can be captured by observable variables is challenged by persistent phenomena, such as volatility clustering, which indicate the influence of unobservable factors shaping market evolution[11].

Definition 2.8 (Latent Variable). A random variable Z_t is called latent at time t if it is $\mathcal{F}_t$-measurable but not $\mathcal{F}_t^{\text{obs}}$-measurable. While Z_t is well-defined on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with a realized value at time t, this value is not accessible given the available information and must therefore be inferred probabilistically.

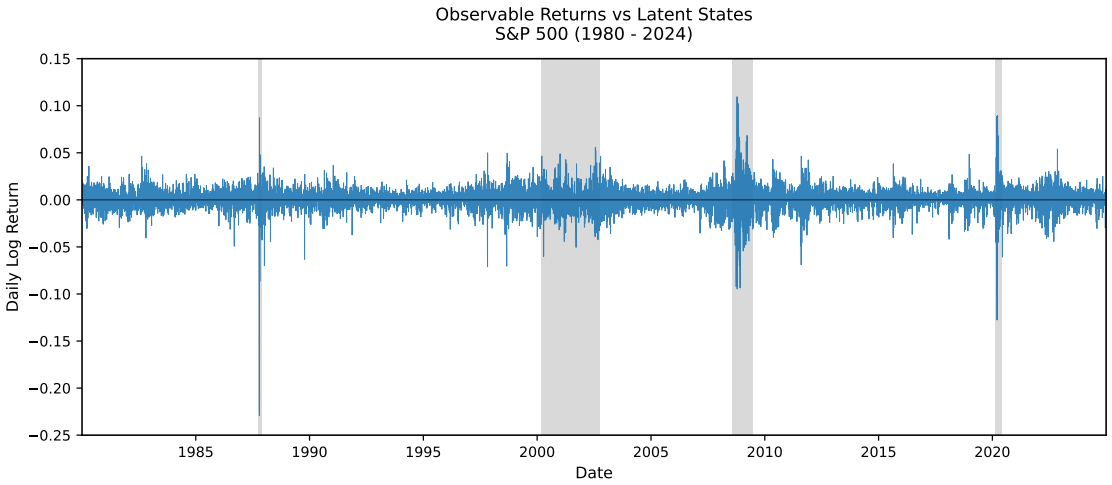


Figure 2.3: *The illustration of partial information highlights the distinction between observable data and latent dynamics. The line represents the logarithmic returns, while the shaded regions mark selected high-volatility episodes used as a visual proxy for latent market conditions not directly observable from the return series alone.*

Consequently, the central mathematical object for dealing with a latent variable is not its exact value, but its posterior distribution conditioned on the observable filtration. If the latent variable Z_t takes values in a state space $\mathcal{Z}$, then for any subset of states $A \subseteq \mathcal{Z}$, this posterior distribution is formally defined as:

$$\pi_t(A) = \mathbb{P}(Z_t \in A \mid \mathcal{F}_t^{\text{obs}}) \tag{2.9}$$

This distribution represents the best possible estimate of the hidden state Z_t given the available market data. When forecasting an observable variable Y_{t+1} influenced by a latent process, we must rely on the tower property of conditional expectation (Eq. 2.4), which provides the appropriate weights to project the latent variables onto the observable information set $\mathcal{F}_t^{\text{obs}}$, accounting for the unobservable parameters:

$$\mathbb{E}[Y_{t+1} \mid \mathcal{F}_t^{\text{obs}}] = \mathbb{E}\Big[\mathbb{E}[Y_{t+1} \mid \mathcal{F}_t] \mid \mathcal{F}_t^{\text{obs}}\Big] \tag{2.10}$$

The inner expectation computes the theoretical dynamics assuming full knowledge of the market's hidden forces, conditional on a realization of the latent variable $Z_t = z$. The outer expectation then integrates out uncertainty, using the posterior distribution $\pi_t(z)$ as a weighting mechanism to average all possible hidden states:

$$\mathbb{E}[Y_{t+1} \mid \mathcal{F}_t^{\text{obs}}] = \int_{\mathcal{Z}} \mathbb{E}[Y_{t+1} \mid \mathcal{F}_t^{\text{obs}} \vee \sigma(Z_t = z)]\, d\pi_t(z) \tag{2.11}$$

This formulation reveals that any forecast made solely on observable data is fundamentally a weighted average. The market's observable behavior is not a unified process, but a dynamic mixture of unobservable components governed by current empirical evidence. To see this in practice, consider the modeling of market risk.

While the logarithmic return r_t is strictly observable and $\mathcal{F}_t^{\text{obs}}$-measurable, the volatility σ_t governing its dispersion is latent. Although fully realized within the global filtration $\mathcal{F}_t$, it remains inaccessible to $\mathcal{F}_t^{\text{obs}}$ as exchanges record only prices, not standard deviations. This partial observability dictates that the variance process resides entirely within $\mathcal{F}_t^{\text{lat}}$, necessitating the probabilistic filtering approach above.

By modeling latent volatility as a **discrete random variable**, the mathematical complexity of the continuous integral over the state space $\mathcal{Z}$ is naturally resolved. This approach simplifies unobservable market forces by grouping them into distinct and persistent phases, providing the foundation for the concept of a *market regime*.

2.3 Market Regime as a Latent Variable

Having established that the fundamental drivers of market risk, such as volatility, reside within the unobservable filtration $\mathcal{F}_t^{\text{lat}}$, we treat them as latent variables. While they could in principle vary continuously over time, modeling them as continuous stochastic processes often leads to significant computational complexity.

To construct a more tractable and interpretable framework, we instead project this spectrum of unobservable forces onto a *finite-dimensional* state space[11].

Definition 2.9 (Discrete Market Regime). A discrete market regime is a latent stochastic process $Z = \{Z_t\}_{t\geq 0}$ taking values in a finite state space:

$$\mathcal{Z} = \{1, 2, \ldots, K\} \tag{2.12}$$

Each state $k \in \mathcal{Z}$ represents a distinct economic regime, such as a calm regime with low variance or a turbulent regime with high variance and determines the parameter vector θ_k, which governs the conditional distribution of observable market data.

This discretization is a structural representation of the empirical stylized facts established in Chapter 1. Specifically, given that the market is in regime $Z_t = k$, we assume that the observable logarithmic return r_t follows a stationary probability density function $f(\cdot)$, parameterized by the state-dependent vector $\theta_k = (\mu_k, \sigma_k^2)$:

$$r_t \mid (Z_t = k, \mathcal{F}_{t-1}^X) \sim f(r_t; \theta_k) \tag{2.13}$$

If the underlying market environment Z_t were known, today's return would be fully described by a stationary distribution determined by that state, conditional on the information available up to $\mathcal{F}_{t-1}^X$. This conditioning on the latent state $Z_t = k$ yields locally stationary dynamics that are consistent with standard Gaussian assumptions.

Because state Z_t is unobserved and time-varying, the observer must integrate over its possible transitions. The resulting switching between parameter vectors θ_k generates the regime-dependent dynamics highlighted in Chapter 1 and the Finite Gaussian Mixture introduced in Definition 1.6. The latent variable $Z_t \in \mathcal{Z}$ selects the active distribution at each time, accounting for both volatility clustering and heavy tails.

The preceding chapters have highlighted the limitations of classical models in capturing the reality of financial markets. Phenomena such as volatility clustering and extreme market shocks are consistent with time-varying market dynamics. By formalizing the flow of information through filtrations and isolating the unobservable risk sources, we project these dynamics onto a tractable discrete latent state space.

While the framework incorporates latent states and conditional distributions, it remains incomplete without specifying their temporal evolution. Modeling the market as a sequence of latent states $\{Z_t\}_{t\geq 0}$ shifts the focus to transition probabilities:

$$p_{ij} = \mathbb{P}(Z_{t+1} = j \mid Z_t = i, \mathcal{F}_t) \tag{2.14}$$

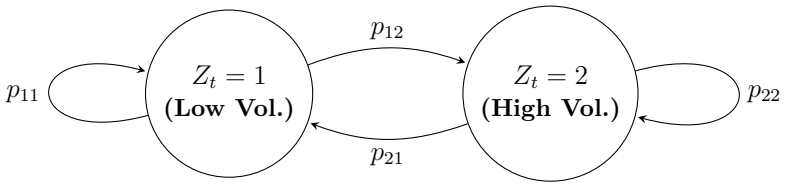


Figure 2.4: *The nodes represent the latent market regimes: low volatility and high volatility, while the directed edges indicate the transition probabilities p_{ij}. This diagram formalizes the dynamics of switching between distinct structural phases.*

To define these transition dynamics and calculate the probabilities of moving between regimes, we must introduce the mathematical machinery of stochastic processes on discrete state spaces. This logical necessity naturally leads our analysis into Part II of this thesis, beginning with the theoretical foundation of Markov Chains.

Combining observable return distributions with the latent Markov transition structure yields a probabilistic framework for modeling markets under partial information. Subsequent chapters address updating the current state given new data, reconstructing past states, and estimating model parameters from observed data*

*The measure-theoretic foundations of probability theory follow Billingsley [8] and Williams [9]. Additionally, Shreve [10] is used for stochastic calculus and Hamilton [11] for unobservable market regimes. Numerical superscripts refer to the complete Bibliography at the end of this work.

Section II

Statistical Models with Discrete and Hidden States

Chapter 3

Markov Chains as a Discrete-State Model

3.1 Fundamental Markov Properties

The Markovian framework provides a representation of state evolution through probabilistic transitions across discrete states, where the conditional distribution of future states depends only on the present state rather than on the entire past history[9]. Widely applied across scientific disciplines, it is highly relevant in quantitative finance for modeling risk dynamics and identifying volatility regimes, as it imposes a highly tractable structure grounded in three core sequential properties.

Definition 3.1 (Finite State Space). Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $T = \{0, 1, 2, \dots\}$ a discrete-time index set. The stochastic process $\{X_t\}_{t \in T}$ takes values in a finite state space $\mathcal{S} = \{s_1, s_2, \dots, s_K\}$, $X_t \in \mathcal{S}$ for all $t \in T$. This assumption ensures the model remains computationally tractable.

Definition 3.2 (Markov Property). A stochastic process $\{X_t\}_{t \in T}$ taking values in a finite state space $\mathcal{S}$ satisfies the Markov property if, for any sequence of states $x_0, x_1, \dots, x_t, x_{t+1} \in \mathcal{S}$, the conditional probability satisfies:

$$\mathbb{P}(X_{t+1} = x_{t+1} \mid X_0 = x_0, \dots, X_t = x_t) = \mathbb{P}(X_{t+1} = x_{t+1} \mid X_t = x_t) \tag{3.1}$$

where $\mathbb{P}(X_0 = x_0, \dots, X_t = x_t) > 0$ ensures the probability is well-defined.

Commonly referred to as *memorylessness*[9], this characteristic dictates that the probability distribution of future states depends exclusively on the **current state**. The sequence of preceding events becomes irrelevant for forecasting, since the state X_t efficiently encapsulates all the necessary data to determine the future trajectory.

Local Conditional Dependence vs. Independence

The operational power of the Markov property lies in replacing pure statistical independence with local conditional dependence. The naive assumption of independence across time can violate basic chronological structure. For example, in the risk analysis of a loan portfolio, a loan evolves through a finite set of delinquency states: *Current, 30–59 Days Late, 60–89 Days Late* and *90+ Days Late*. If monthly observations were assumed to be independent, a loan could move from *Current* to *90+ Days Late* in a single step, an outcome that is chronologically impossible yet would still receive non-zero probability under a naive independent model.

A Markovian structure resolves this by enforcing **first-order Markov dependence:** the probability of the next state is determined solely by the current one. In the loan example, this encodes chronological constraints directly into the model. A loan in the *Current* state is assigned zero probability of transitioning immediately to *90+ Days Late*, while a loan already in *90+ Days Late* has a high probability of remaining there. By conditioning only on the present state, the Markov chain rules out impossible transitions and captures the persistence typical of financial variables.

Definition 3.3 (Time-Homogeneity). A Markov chain is considered time-homogeneous if the conditional probability of moving from state i to state j is invariant with respect to the specific time index t[9]. Mathematically, for any $t \in T$ and $i, j \in \mathcal{S}$:

$$\mathbb{P}(X_{t+1} = j \mid X_t = i) = p_{ij}. \tag{3.2}$$

This assumption requires that, although the process evolves dynamically, the one-step transition probabilities remain constant. Therefore, a time-homogeneous discrete-time Markov chain is specified by the triplet $(\mathcal{S}, P, \alpha)$, where $\mathcal{S}$ is the finite state space, $P = (p_{ij})$ is the transition matrix and α is the initial distribution, with $\alpha_i = \mathbb{P}(X_0 = i)$ for each $i \in \mathcal{S}$ and $\sum_{i \in \mathcal{S}} \alpha_i = 1$, allowing multi-step computations.

3.2 Transition Matrix, Stationary Distribution

The dynamics of a time-homogeneous Markov chain over a finite state space $\mathcal{S} = \{s_1, s_2, \ldots, s_K\}$ are mathematically governed by its transition probabilities. The conditional probability of migrating from state i to state j in a single time step is denoted by p_{ij}. These probabilities are organized into a $K \times K$ transition matrix P:

$$P = \begin{pmatrix} p_{11} & p_{12} & \cdots & p_{1K} \\ p_{21} & p_{22} & \cdots & p_{2K} \\ \vdots & \vdots & \ddots & \vdots \\ p_{K1} & p_{K2} & \cdots & p_{KK} \end{pmatrix} \tag{3.3}$$

As it represents probabilities, the transition matrix must satisfy two constraints[12]:

- All entries must be non-negative, $p_{ij} \geq 0$, since probabilities are non-negative;
- Each row must sum to one, $\sum_{j=1}^{K} p_{ij} = 1$, ensuring a valid distribution;

Consider the transition matrix associated with the loan portfolio introduced in the previous section. Let the state space be defined as $\mathcal{S} = \{1,2,3,4\}$, corresponding to the delinquency states *Current*, *30–59 Days Late*, *60–89 Days Late*, and *90+ Days Late*, respectively. The transition matrix P for the monthly credit cycle reflects the local dependence structure, where admissible transitions are constrained by the chronological ordering of delinquency states. In particular, transitions that skip intermediate states within a single period are excluded, implying $p_{13} = p_{14} = p_{24} = 0$:

$$P = \begin{pmatrix} p_{11} & p_{12} & 0 & 0 \\ p_{21} & p_{22} & p_{23} & 0 \\ p_{31} & p_{32} & p_{33} & p_{34} \\ p_{41} & p_{42} & p_{43} & p_{44} \end{pmatrix}$$

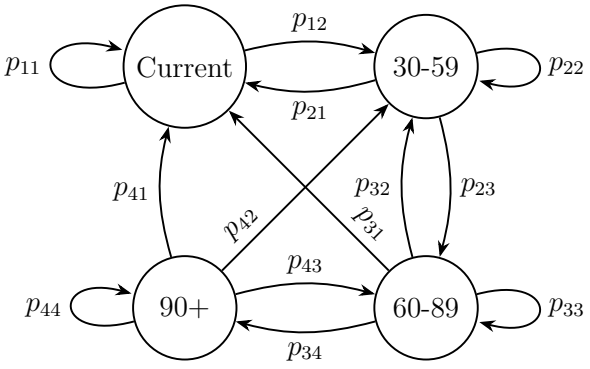


Figure 3.1: *Algebraic and topological representation of local conditional dependence.*

While the transition matrix P governs single-step movements, financial risk management requires projections over extended horizons. Let α_t be a $1 \times K$ row vector representing the state probability distribution at time t, where the i-th component is $\alpha_t(i) = \mathbb{P}(X_t = i)$. The temporal evolution of this distribution is, then, given by:

$$\alpha_{t+1} = \alpha_t P \tag{3.4}$$

The distribution at future horizon n is obtained through the n-step transition matrix:

$$\alpha_n = \alpha_0 P^n \tag{3.5}$$

Given monthly frequency of the credit cycle, the distribution after one year ($n = 12$) is computed as $\alpha_{12} = \alpha_0 P^{12}$, where the fourth element of the resulting vector is the proportion of loans expected in the 90+ Days Late state at the end of the year.

The long-run behavior of a Markov chain is characterized by a **stationary distribution**. It is important to distinguish it from the existence of a limiting distribution, since stationarity, uniqueness and convergence are different mathematical properties.

Definition 3.4 (Stationary Distribution). A probability vector is stationary if[9]:

$$\pi P = \pi, \qquad \sum_{i \in \mathcal{S}} \pi_i = 1, \qquad \pi_i \geq 0 \quad \forall i \in \mathcal{S} \tag{3.6}$$

If the chain starts with initial distribution $\alpha_0 = \pi$, then the distribution does not change over time, since $\alpha_t = \pi$ for every $t \geq 0$. To state when convergence toward a unique distribution occurs, some structural properties must first be introduced[9]:

- **Irreducibility:** every state can eventually be reached from any other state. For any $i, j \in \mathcal{S}$, there exists at least one integer $n \geq 0$ such that $p_{ij}^{(n)} > 0$.
- **Positive recurrence:** starting from a given state i, the expected time needed to return to that state is finite. If $T_i^+ = \inf\{n \geq 1 : X_n = i\}$ denotes the first return time to state i, positive recurrence is expressed as $\mathbb{E}_i[T_i^+] < \infty$.
- **Aperiodicity:** the chain avoids fixed-length return cycles. An irreducible chain is aperiodic if the greatest common divisor of return times for its states is equal to one: $d(i) = \gcd\{n \geq 1 : p_{ii}^{(n)} > 0\} = 1$.

Definition 3.5 (Limiting Distribution). A probability vector π is a limiting distribution if the chain converges to π independently of the initial distribution α_0:

$$\lim_{n\to\infty} \alpha_0 P^n = \pi, \qquad \forall \alpha_0 \text{ probability distribution on } \mathcal{S}. \tag{3.7}$$

Theorem 3.1 (Convergence to Equilibrium). If a time-homogeneous Markov chain is irreducible, positive recurrent and aperiodic, then it admits a unique stationary distribution π, which corresponds also to the limiting distribution[9]:

$$\lim_{n\to\infty} p_{ij}^{(n)} = \pi_j \qquad \forall i, j \in \mathcal{S}. \tag{3.8}$$

In the finite-state case considered in this thesis, irreducibility implies positive recurrence. Therefore, for a finite, irreducible, and aperiodic Markov chain, $P^n \longrightarrow \mathbf{1}\pi$, where $\mathbf{1}$ is a column vector of ones and each row of the limiting matrix equals π.

Under these conditions, the system is ergodic: the influence of α_0 vanishes as $n \to \infty$. If the chain is reducible or contains absorbing classes, the limiting behavior may depend on the initial distribution and a unique global equilibrium may fail to exist.

The stationary condition is linked to linear algebra: since time-homogeneity makes P fixed, the equilibrium condition can be studied through its spectral properties[12].

Definition 3.6 (Eigenvector and Eigenvalue). Let A be a square matrix. A non-zero row vector v is a left eigenvector of A, associated with eigenvalue λ, if[12]:

$$vA = \lambda v. \tag{3.9}$$

Applying this definition to the transition matrix, the stationary condition becomes

$$\pi P = 1 \cdot \pi. \tag{3.10}$$

Therefore, π is the left eigenvector of P associated with eigenvalue 1. For irreducible chains, this stationary distribution is unique. However, in reducible chains, multiple stationary distributions may exist and the behavior can depend on the initial state.

Example: Computing a Stationary Distribution

Consider a simple two-state Markov chain with transition matrix:

$$P = \begin{pmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{pmatrix}, \qquad \pi = (\pi_1, \pi_2).$$

Recalling Eqs. (3.9)–(3.10), the stationary distribution must satisfy:

$$\pi P = \pi, \qquad \pi_1 + \pi_2 = 1.$$

Expanding the matrix product:

$$(\pi_1, \pi_2) \begin{pmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{pmatrix} = (0.8\pi_1 + 0.3\pi_2,\ 0.2\pi_1 + 0.7\pi_2).$$

$$0.8\pi_1 + 0.3\pi_2 = \pi_1, \qquad 0.2\pi_1 + 0.7\pi_2 = \pi_2,$$

$$0.2\pi_1 = 0.3\pi_2 \qquad \Longrightarrow \qquad \pi_2 = \frac{2}{3}\pi_1.$$

Using the normalization condition:

$$\pi_1 + \frac{2}{3}\pi_1 = 1 \qquad \Longrightarrow \qquad \pi_1 = \frac{3}{5} = 0.6, \qquad \pi_2 = \frac{2}{5} = 0.4.$$

$$\boxed{\pi = (0.6{,}0.4)}.$$

The stationary condition can be verified directly:

$$(0.6{,}0.4) \begin{pmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{pmatrix} = (0.48 + 0.12,\ 0.12 + 0.28) = (0.6{,}0.4).$$

Therefore, under $\pi = (0.6{,}0.4)$, one transition leaves the state probabilities unchanged, illustrating stationarity as the invariance of the distribution under P.

3.3 The Chapman-Kolmogorov Equations

While spectral analysis identifies the asymptotic destination, financial risk management requires the exact probability flow across finite time intervals. The theoretical framework for this operation is the Chapman-Kolmogorov equations, which formalize how long-term transitions can be decomposed into sequences of intermediate steps.

Definition 3.7 (n-step Transition Probability). The n-step transition probability, denoted by $p_{ij}^{(n)}$, is defined as the conditional probability that a system currently residing in state i will be found in state j after exactly n transition steps[9]:

$$p_{ij}^{(n)} = \mathbb{P}(X_n = j \mid X_0 = i) \tag{3.11}$$

where $p_{ij}^{(1)} = p_{ij}$ represents the standard single-step transition.

Theorem 3.2 (Chapman-Kolmogorov). For any states $i, j \in \mathcal{S}$ and any non-negative integers n, m, the Chapman-Kolmogorov equation formally states that[9, 11]:

$$p_{ij}^{(n+m)} = \sum_{k \in \mathcal{S}} p_{ik}^{(n)} p_{kj}^{(m)} \tag{3.12}$$

To transition from an initial state i to a target state j over $n+m$ discrete steps, the process must pass through the intermediate state $k \in \mathcal{S}$ at time n. Since the states are mutually exclusive, the total probability is the sum over all intermediate states.

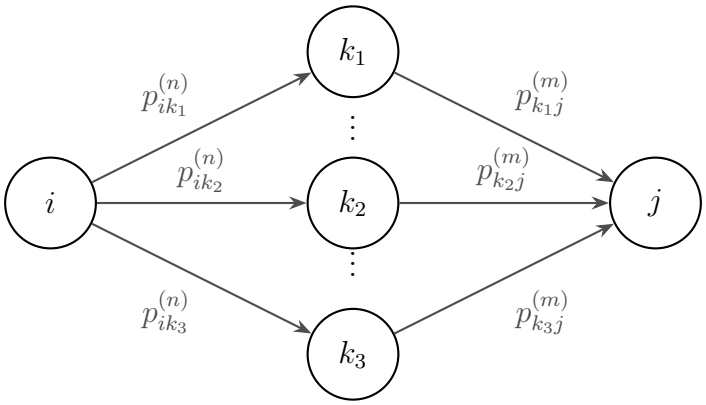


Figure 3.2: *The $(n+m)$-step transition probability is obtained by marginalizing over the intermediate state at time n, summing over $k \in \mathcal{S}$ the product of the n-step and m-step transition probabilities, a factorization justified by the Markov property.*

To derive this identity, the state trajectory is partitioned into two sequential phases at an intermediate step n. By applying the *Law of Total Probability*, the transition is expressed as a sum over the state space $\mathcal{S}$ within a single expression:

$$p_{ij}^{(n+m)} = \mathbb{P}(X_{n+m} = j \mid X_0 = i) = \sum_{k \in \mathcal{S}} \mathbb{P}(X_{n+m} = j, X_n = k \mid X_0 = i)$$

The joint conditional probability is then factorized using the *chain rule*, $\mathbb{P}(A, B \mid C) = \mathbb{P}(A \mid B, C)\mathbb{P}(B \mid C)$, where A and B denote the states at times $n+m$ and n, respectively, with the factors ordered to reflect the transition from i to k to j:

$$= \sum_{k \in \mathcal{S}} \mathbb{P}(X_{n+m} = j \mid X_n = k, X_0 = i) \cdot \mathbb{P}(X_n = k \mid X_0 = i)$$

Finally, by the *Markov property*, the future evolution after time n depends only on the intermediate state $X_n = k$. Therefore, the first conditional factor becomes:

$$\mathbb{P}(X_{n+m} = j \mid X_n = k, X_0 = i) = \mathbb{P}(X_{n+m} = j \mid X_n = k) = p_{kj}^{(m)}. \tag{3.13}$$

The second factor is the probability of reaching the intermediate state k from i:

$$\mathbb{P}(X_n = k \mid X_0 = i) = p_{ik}^{(n)}. \tag{3.14}$$

$$p_{ij}^{(n+m)} = \sum_{k \in \mathcal{S}} p_{ik}^{(n)} p_{kj}^{(m)}. \tag{3.15}$$

Algebraically, the summation $\sum_{k \in \mathcal{S}} p_{ik}^{(n)} p_{kj}^{(m)}$ corresponds to the inner product between the i-th row of $P^{(n)}$ and the j-th column of $P^{(m)}$. This, eventually, validates the equivalence between multi-step transition probabilities and matrix multiplication:

$$P^{(n+m)} = P^{(n)} P^{(m)}. \tag{3.16}$$

The n-step transition matrix $P^{(n)}$ is the n-th power of the one-step matrix P:

$$P^{(n)} = P^n. \tag{3.17}$$

This result justifies the matrix operator in Section 3.2, ensuring that forecasting the loan portfolio over annual horizon through P^{12} accounts for all intermediate states.

Example: Computing Multi-Step Transition Probabilities

Consider the same two-state Markov chain introduced in Section 3.2:

$$P = \begin{pmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{pmatrix}$$

Recalling the Chapman–Kolmogorov relation in Eq. (3.12), the two-step transition matrix is obtained by composing two consecutive one-step transitions:

$$P^{(2)} = P^2 = P \cdot P$$

Expanding the matrix product:

$$P^2 = \begin{pmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{pmatrix} \begin{pmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{pmatrix}$$

$$= \begin{pmatrix} 0.8(0.8) + 0.2(0.3) & 0.8(0.2) + 0.2(0.7) \\ 0.3(0.8) + 0.7(0.3) & 0.3(0.2) + 0.7(0.7) \end{pmatrix}$$

$$= \begin{pmatrix} 0.70 & 0.30 \\ 0.45 & 0.55 \end{pmatrix}$$

For example, the probability of moving from state 1 to state 2 in two steps is:

$$p_{12}^{(2)} = p_{11}p_{12} + p_{12}p_{22} = 0.8(0.2) + 0.2(0.7) = 0.30.$$

The two terms represent the two possible intermediate paths:

$$1 \to 1 \to 2 \qquad \text{and} \qquad 1 \to 2 \to 2.$$

Thus, the two-step transition probability is obtained by summing the probabilities of all possible admissible paths through each of the intermediate states, providing a direct numerical illustration of the Chapman–Kolmogorov equation.

3.4 Maximum Likelihood Estimation of Transition Probabilities

While Markov chains provide a rigorous framework for state evolution, their application relies on estimating the transition matrix P from observed data[12]. The standard approach to infer these parameters is the *Maximum Likelihood Estimation* (MLE), which determines the values that maximize the likelihood of the sample[12, 13].

Definition 3.8 (Trajectory Likelihood). Let $(X_t)_{t=0}^n$ be a Markov chain with transition matrix P, and let $x_0, x_1, \ldots, x_n$ be an observed sequence. The likelihood function $L(P)$ is the joint probability of observing the sequence in the given order:

$$L(P) = \mathbb{P}(X_0 = x_0, X_1 = x_1, \ldots, X_n = x_n) \tag{3.18}$$

Evaluating this joint probability for a long sequence is intractable. Using the *chain rule of probability*, it can first be written as a product of conditional probabilities. The *Markov property* then eliminates the dependence on the full history, yielding:

$$L(P) = \mathbb{P}(X_0 = x_0) \prod_{t=1}^{n} \mathbb{P}(X_t = x_t \mid X_{t-1} = x_{t-1}) = \mathbb{P}(X_0 = x_0) \prod_{t=1}^{n} p_{x_{t-1}x_t} \tag{3.19}$$

This factorization highlights that, in Markov models, a full trajectory reduces to a simple product of single-step transition probabilities, simplifying the computation. In terms of observed transition frequencies, we first define the **transition counts:**

$$n_{ij} = \sum_{t=1}^{n} \mathbf{1}_{\{x_{t-1}=i,\, x_t=j\}}, \qquad n_i = \sum_{j \in S} n_{ij}. \tag{3.20}$$

where n_{ij} is the number of observed transitions from i to j and n_i is the total number of transitions leaving state i. The likelihood function can be rewritten as:

$$L(P) = \mathbb{P}(X_0 = x_0) \prod_{i \in \mathcal{S}} \prod_{j \in \mathcal{S}} p_{ij}^{n_{ij}} \tag{3.21}$$

For computational optimization, it is standard to compute the log-likelihood $\ell(P) = \ln L(P)$, which transforms the multiplicative structure into a linear summation:

$$\ell(P) = \ln \mathbb{P}(X_0 = x_0) + \sum_{i \in \mathcal{S}} \sum_{j \in \mathcal{S}} n_{ij} \ln p_{ij} \tag{3.22}$$

Consider a sample given by the sequence $1 \to 2 \to 3 \to 1 \to 2$. The joint probability of this path is $\mathbb{P}(X_0 = 1) \cdot p_{12} \cdot p_{23} \cdot p_{31} \cdot p_{12}$. Using transition counts, it simplifies to $p_{12}^2 \cdot p_{23} \cdot p_{31}$, which under a logarithmic transformation becomes $2 \ln p_{12} + \ln p_{23} + \ln p_{31}$.

The next objective is to determine the optimal parameter values p_{ij} that maximize this function, strictly adhering to the axiomatic properties of Markovian systems.

Definition 3.9 (Row-Stochastic Transition Matrix). The transition matrix P of any discrete-time Markov chain must be row-stochastic, meaning that[12]:

$$\sum_{j \in S} p_{ij} = 1, \qquad p_{ij} \geq 0, \qquad \forall i, j \in S \tag{3.23}$$

To solve this maximization, we employ the Lagrange multipliers[12]. By introducing a set of multipliers λ_i for each state i, we formulate the Lagrangian function Λ:

$$\Lambda(P, \lambda) = \sum_{i \in \mathcal{S}} \sum_{j \in \mathcal{S}} n_{ij} \ln p_{ij} - \sum_{i \in \mathcal{S}} \lambda_i \left(\sum_{j \in \mathcal{S}} p_{ij} - 1 \right) \tag{3.24}$$

The partial derivative with respect to a generic parameter p_{ij} set to zero yields:

$$\frac{\partial \Lambda}{\partial p_{ij}} = \frac{n_{ij}}{p_{ij}} - \lambda_i = 0 \implies p_{ij} = \frac{n_{ij}}{\lambda_i} \tag{3.25}$$

By substituting this expression back into the row-sum constraint, we find that $\sum_j \frac{n_{ij}}{\lambda_i} = 1$, which implies that $\lambda_i = \sum_j n_{ij}$. Consequently, the MLE for the transition probability is simply the relative frequency of the observed transitions:

$$\hat{p}_{ij} = \frac{n_{ij}}{\sum_{k \in \mathcal{S}} n_{ik}} \tag{3.26}$$

To bridge this theoretical framework, the following application shows how the transition counts are extracted from raw data and transformed into the matrix $\hat{P}$.

Example: Calibrating the Transition Matrix

First, we illustrate how the transition counts n_{ij} are obtained from observed data. Consider a simplified portfolio of three independent loans observed over four months, with state space $\mathcal{S} = \{1 \textit{ (On time)}, 2 \textit{ (Late payment)}, 3 \textit{ (Severely late)}\}$:

$$\textbf{Loan A: } 1 \to 1 \to 1 \to 2$$
$$\textbf{Loan B: } 1 \to 2 \to 1 \to 1$$
$$\textbf{Loan C: } 2 \to 3 \to 2 \to 3$$

The transition counts n_{ij} are obtained by recording each consecutive pair (x_{t-1}, x_t) across the trajectories. For example, $1 \to 1$ occurs twice in Loan A and once in Loan B, so $n_{11} = 3$, while $1 \to 2$ occurs once in each, giving $n_{12} = 2$. Aggregating all step-by-step transitions constructs the absolute count matrix N:

$$N = \begin{bmatrix} n_{11} & n_{12} & n_{13} \\ n_{21} & n_{22} & n_{23} \\ n_{31} & n_{32} & n_{33} \end{bmatrix} = \begin{bmatrix} 3 & 2 & 0 \\ 1 & 0 & 2 \\ 0 & 1 & 0 \end{bmatrix} \tag{3.27}$$

The derived Maximum Likelihood Estimator $\hat{p}_{ij} = n_{ij} / \sum_k n_{ik}$ is now applied to normalize the matrix. Each element is divided by its respective row sum. This structural operation yields the final calibrated transition probability matrix $\hat{P}$:

$$\hat{P} = \begin{bmatrix} 3/5 & 2/5 & 0/5 \\ 1/3 & 0/3 & 2/3 \\ 0/1 & 1/1 & 0/1 \end{bmatrix} \approx \begin{bmatrix} 0.60 & 0.40 & 0.00 \\ 0.33 & 0.00 & 0.67 \\ 0.00 & 1.00 & 0.00 \end{bmatrix} \tag{3.28}$$

This numerical demonstration confirms that the theoretical MLE naturally enforces the fundamental probability axioms, ensuring that each row strictly sums to unity (1.00) while reflecting the observed migration frequencies of the portfolio.

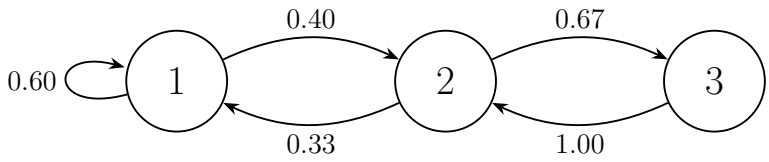


Figure 3.3: *Topological representation of the empirically calibrated Markov chain.*

From Observable States to Hidden Regimes

The mathematical framework developed in this chapter provides a basis for modelling stochastic evolution. However, the validity of the standard Markov model relies on a strong assumption: the state space $\mathcal{S}$ is fully observable and the system's state can be identified at any time t. In the credit risk example, this reflects an exact correspondence between the risk classification and the client's financial condition.

In the real financial world, this assumption of perfect observability does not hold as the true underlying state is latent. Analysts can observe only probabilistic signals, such as delayed transactions or fluctuating credit scores. Treating these observations as true can introduce structural bias and lead to inaccurate risk assessments.

Therefore, the framework must be extended to a *hidden-state model*, leading to the theory of **Hidden Markov Models (HMMs)**[11, 13]. In an HMM, the system evolves according to an unobservable Markov process, while a second stochastic layer connects hidden states to noisy observations. Since the true states are latent, the frequency-based estimator is inapplicable. Chapter 4 will present the recursive algorithms used to infer hidden regimes, enabling probabilistic regime inference.*

*The theoretical foundations of Markov chains and their estimation follow Williams [9] and Boas [12]. Additionally, Hamilton [11], Guidolin [13] and Krolzig [14] are used for hidden market states. Numerical superscripts refer to the complete Bibliography at the end of this work.

Chapter 4

Hidden Markov Models: Theory and Inference

4.1 Discrete States and Continuous Emissions

The previous chapter analyzed the properties of Markovian processes under the assumption of perfect observability, where the state X_t is directly observable and fully reflected in the information set $\mathcal{F}_t$. In reality, however, financial phenomena often exhibit structural *latency*: we are able to observe market outputs, such as price changes or volatility clusters, while the generative regime remains hidden.

The Hidden Markov Model (HMM) addresses this gap by introducing a double stochastic process[15, 18]. The first is a **latent Markov chain** $\{X_t\}_{t\in T}$, taking values in a finite state space, while the second is an **observable return process** $\{r_t\}_{t\in T}$ generated by the hidden state. In this interpretation, the market regime remains discrete and latent, whereas the observed logarithmic return is continuous.

Definition 4.1 (The Latent Process). Let $\{X_t\}_{t\in T}$ be a time-homogeneous Markov chain taking values in a finite state space $\mathcal{S} = \{s_1, s_2, \ldots, s_K\}$, with transitions governed by the matrix $P = (p_{ij})$, where $p_{ij} = \mathbb{P}(X_{t+1} = s_j \mid X_t = s_i)$. In an HMM structure, this sequence of states is strictly latent. Therefore, the regime evolution must be inferred probabilistically from the observable return sequence $\{r_t\}_{t\in T}$.

In the credit models of Chapter 3, a loan's delinquency status is recorded. For the HMM framework, consider modeling a firm's *financial health*, an unobservable

condition. The challenge lies in decoding whether a negative return is a transition to a *distressed* regime or merely stochastic noise within a *healthy* state. The mapping from the hidden regime to the observed return is formalized by the emission density.

Definition 4.2 (State-Dependent Gaussian Emission). Let $r_t \in \mathbb{R}$ denote the observed logarithmic return at time t. The observation process is coupled with the latent chain through state-dependent emission densities $b_j(r)$, where each density describes the distribution of returns conditional on the hidden state $X_t = s_j$[17, 18]:

$$b_j(r) = f(r; \theta_j), \qquad 1 \leq j \leq K. \tag{4.1}$$

In financial Gaussian HMMs, state s_j is parameterized by $\theta_j = (\mu_j, \sigma_j^2)$, such that:

$$r_t \mid X_t = s_j \sim \mathcal{N}(\mu_j, \sigma_j^2). \tag{4.2}$$

Each hidden state defines a local return distribution. A low-volatility regime exhibits a smaller variance σ_j^2, while a turbulent regime exhibits a larger variance.

This formulation recalls the finite Gaussian mixture introduced in Definition 1.6, where the unconditional distribution of returns is represented as a combination of Gaussian components. The crucial difference is dynamic: in a static mixture, the component weights are fixed and independent across time, whereas in an HMM the active component is selected by the latent Markov chain $\{X_t\}_{t \in T}$. The transition matrix P therefore transforms a static mixture of distributions into a regime-switching model capable of capturing persistence and volatility clustering.

To illustrate, assume a firm operates under $K = 2$ regimes ($s_1 =$ Healthy, $s_2 =$ Distressed). Under s_1, returns fluctuate around a stable mean with low dispersion, while under distress they exhibit larger negative movements and higher volatility:

$$r_t \mid X_t = s_1 \sim \mathcal{N}(\mu_1, \sigma_1^2), \qquad r_t \mid X_t = s_2 \sim \mathcal{N}(\mu_2, \sigma_2^2), \qquad \sigma_1^2 < \sigma_2^2.$$

A large negative return is more compatible with the distressed regime, but it is not sufficient to identify it with certainty. The true state remains strictly hidden because the observed return is only a noisy signal generated by the underlying regime.

Definition 4.3 (The HMM Triplet). A Gaussian Hidden Markov Model is fully parameterized by the triplet $\lambda = (P, B, \alpha)$, where P is the $K \times K$ transition matrix governing the latent dynamics, $B = \{b_j(\cdot)\}_{j=1}^{K}$ is the family of state-dependent emission densities, and α is the initial state distribution, with $\alpha_i = \mathbb{P}(X_1 = s_i)$[16, 17].

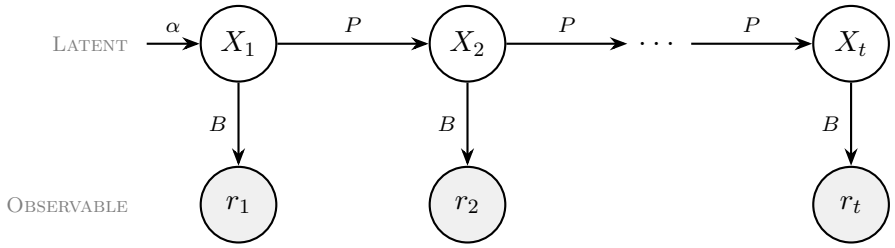


Figure 4.1: *The distribution α initializes the latent sequence, P propagates the hidden transitions, and B determines the density generating the return r_t.*

The tractability of the model λ relies on two structural independence assumptions: (i) *First-Order Markovianity*, where the latent sequence $\{X_t\}$ satisfies the Markov property defined in Section 3.1; and (ii) *Local Conditional Independence*, which dictates that the conditional distribution of the current return r_t is determined by the hidden state X_t, rendering it independent of all past states and observations[17]:

$$p(r_t \mid X_1, \ldots, X_t, r_1, \ldots, r_{t-1}) = p(r_t \mid X_t). \tag{4.3}$$

Under this architecture, the latent regime acts as a return generator. Here, the challenge reduces to an inverse problem: given a recorded return sequence $r_{1:T}$, the observer must statistically infer the most likely underlying trajectory $X_{1:T}$[18].

4.2 The Three Fundamental Problems

While the triplet $\lambda = (P, B, \alpha)$ defines the architecture of the model, its utility depends on the ability to extract meaningful information from observations. Defining the state-dependent emission law B is *necessary but not sufficient*: **Rabiner**'s framework[18] categorizes the application of HMMs into three core problems[15, 17].

Let $r_{1:T} = \{r_1, r_2, \ldots, r_T\}$ be an observed sequence of T logarithmic returns. To make the HMM structure viable, the following three challenges must be resolved.

Problem 1: Model Evaluation and Selection

The first computational challenge is computing the marginal likelihood $p(r_{1:T} \mid \lambda)$ of an observed return sequence $r_{1:T}$, given a fully parameterized model λ. In probability theory, marginalization is the process of isolating the likelihood of the observable data by averaging over all possible hidden states. As anticipated in Section 2.2, this translates to summing the joint likelihood over every hidden trajectory $X_{1:T}$:

$$p(r_{1:T} \mid \lambda) = \sum_{\text{all } X} p(r_{1:T} \mid X, \lambda)\mathbb{P}(X \mid \lambda). \tag{4.4}$$

This problem is essential for model assessment, as computing the exact likelihood provides an objective metric of how well a model describes the observed data. It allows one to compare different structures, such as a *2-regime versus 3-regime* model.

Problem 2: State Inference

The second challenge focuses on uncovering the hidden dynamics by inferring the *optimal* sequence of latent states $X_{1:T}$ that most likely generated the observed returns $r_{1:T}$, assuming the model λ is known. *Optimality* follows two approaches:

- **Local Decoding:** seeks the most likely hidden state at each time step t, without requiring the whole sequence to be jointly optimal. For each t we choose the state with the highest posterior probability: $\hat{x}_t = \arg\max_i \mathbb{P}(X_t = s_i \mid r_{1:T}, \lambda)$.
- **Global Decoding:** seeks the most likely complete sequence of hidden states, considering the trajectory as a whole: $x^*_{1:T} = \arg\max_{x_{1:T}} \mathbb{P}(x_{1:T} \mid r_{1:T}, \lambda)$.

Concatenating locally optimal states does not inherently yield the globally optimal path. A sequence of individually probable states might be structurally impossible if the transition matrix P assigns it a zero probability. This limitation of local inference is resolved by global decoding, which enforces Markovian continuity across the entire trajectory through the *Viterbi algorithm*, later derived in Section 4.3.

This difference dictates their distinct applications: *local decoding* is useful for assigning state probabilities at each point in time, whereas *global decoding* facilitates backtesting by reconstructing the most likely historical sequence of past regimes.

Problem 3: Parameter Calibration

The third challenge addresses the calibration of the model, which requires optimizing the parameter triplet $\lambda = (P, B, \alpha)$ to maximize the likelihood of the observed return sequence $p(r_{1:T} \mid \lambda)$. Unlike the fully observable Markov chains, where MLE reduces to simple transition counting, the HMM framework lacks the realized true states.

This forces a summation over all possible trajectories *inside* the logarithm $\ell(\lambda)$:

$$\ell(\lambda) = \log p(r_{1:T} \mid \lambda) = \log \left(\sum_{\text{all } X} p(r_{1:T} \mid X, \lambda) \mathbb{P}(X \mid \lambda) \right) \tag{4.5}$$

Because the logarithm of a sum cannot be decomposed to solve $\nabla_\lambda \ell(\lambda) = 0$, a closed-form solution is unavailable. This intractability is addressed through the *Baum-Welch algorithm*, detailed in Section 4.4. Resolving this problem enables the model to fit a statistical representation of underlying market dynamics.

The mathematical obstacle linking these three problems is the **computational complexity** generated by the hidden state space. A naive approach to Problem 1 or Problem 2 requires evaluating all K^T possible hidden trajectories, resulting in a complexity of order $\mathcal{O}(TK^T)$. In a standard financial dataset with $T = 100$ trading days and $K = 2$ regimes, this already yields $2^{100} \approx 1.26 \times 10^{30}$ possible paths.

The operational implementation of HMMs therefore relies on dynamic programming algorithms. By exploiting the Markov property and local conditional independence, the *Forward, Viterbi and Baum-Welch* recursions reduce the computational complexity to $\mathcal{O}(K^2 T)$, linear in the time and quadratic in the number of states.

This justifies the introduction of the Forward, Viterbi and Baum-Welch algorithms, derived in the subsequent sections as the computational solutions to Problems 1, 2 and 3. Forward and Viterbi provide exact recursive solutions for fixed model parameters, while Baum-Welch provides an iterative estimation procedure.

4.3 Inference via Forward-Backward and Viterbi

This section details techniques to solve Problems 1 and 2. By propagating partial computations along time series, Forward, Forward-Backward and Viterbi recursions transform intractable hidden-state summations into step-by-step iterations[17].

The Forward Algorithm

This algorithm[18] computes the marginal likelihood of an observed sequence in an HMM. In the financial setting, this means evaluating how likely the observed return sequence is under a given model λ. A direct computation would require considering every possible sequence of hidden regimes and summing their likelihood contributions. This computational issue is avoided by exploiting the Markov property: instead of evaluating all hidden paths separately, at each time t, the algorithm evaluates the joint likelihood of observing the sequence up to that time and being in a specific hidden state, then passes these quantities *forward* to the next time step.

Definition 4.4 (The Forward Variable). Given an HMM with $\lambda = (P, B, \alpha)$, the forward variable $\varphi_t(i)$ denotes the joint likelihood of observing the partial return sequence $r_{1:t} = (r_1, r_2, \ldots, r_t)$ and occupying the latent state s_i at time t:

$$\varphi_t(i) \triangleq p(r_{1:t}, X_t = s_i \mid \lambda). \tag{4.6}$$

where X_t denotes the hidden state at time t, while the symbol $\triangleq$ means *is defined as*. The marginal likelihood of the return sequence, $p(r_{1:T} \mid \lambda)$, is evaluated through:

1. **Initialization** ($t = 1$): for each hidden state $s_i \in \mathcal{S}$, the joint likelihood of starting in state s_i and observing the first return r_1 is computed by combining the initial state probability α_i with the emission density $b_i(r_1)$:

$$\varphi_1(i) = \alpha_i b_i(r_1). \tag{4.7}$$

2. **Induction** ($t = 2, 3, \ldots, T$): for each hidden state $s_j \in \mathcal{S}$, the algorithm computes the likelihood of reaching state s_j at time t after having observed the return sequence up to time $t - 1$, by summing over all possible previous states

s_i, weighted by p_{ij}. The resulting quantity is then multiplied by the emission density $b_j(r_t)$ of observing the current return r_t from state s_j:

$$\varphi_t(j) = \left(\sum_{i=1}^{K} \varphi_{t-1}(i) p_{ij} \right) b_j(r_t). \tag{4.8}$$

3. **Termination**: in step T, the likelihood of the observed return sequence is obtained by summing the final forward variables over all possible hidden states:

$$p(r_{1:T} \mid \lambda) = \sum_{i=1}^{K} \varphi_T(i). \tag{4.9}$$

Application: Forward Algorithm with Gaussian Emissions

Consider a Gaussian HMM with two latent regimes, representing the unobservable financial condition of the market at time t: $\mathcal{S} = \{s_1, s_2\}$, where $s_1 =$ Stable and $s_2 =$ Distressed. The observed variables are logarithmic returns: $r_{1:2} = (-0.035, -0.028)$.

- **Initial Distribution:**

$$\boldsymbol{\alpha} = \begin{bmatrix} 0.6 & 0.4 \end{bmatrix}$$

- **Transition Matrix:**

$$P = \begin{bmatrix} 0.85 & 0.15 \\ 0.25 & 0.75 \end{bmatrix}$$

- **Gaussian Emission Densities:**

$$r_t \mid X_t = s_1 \sim \mathcal{N}(0{,}0.01^2), \qquad r_t \mid X_t = s_2 \sim \mathcal{N}(-0.01{,}0.03^2)$$

Evaluating the Gaussian densities at the two observed returns gives:

$$b_1(r_1) = 0.0873, \qquad b_2(r_1) = 9.3971,$$

$$b_1(r_2) = 0.7915, \qquad b_2(r_2) = 11.1075.$$

The objective is to compute the likelihood $p(r_{1:2} \mid \lambda)$.

1. **Initialization** $(t = 1)$:

$$\varphi_1(s_1) = \alpha_1 b_1(r_1) = 0.6 \cdot 0.0873 = 0.0524,$$
$$\varphi_1(s_2) = \alpha_2 b_2(r_1) = 0.4 \cdot 9.3971 = 3.7588.$$

2. **Induction** $(t = 2)$:

   For the *stable* regime:

$$\begin{aligned}\varphi_2(s_1) &= \left[\varphi_1(s_1)p_{11} + \varphi_1(s_2)p_{21}\right] b_1(r_2) \\ &= \left[0.0524(0.85) + 3.7588(0.25)\right] 0.7915 \\ &= \left[0.0445 + 0.9397\right] 0.7915 \\ &= 0.7790.\end{aligned}$$

   For the *distressed* regime:

$$\begin{aligned}\varphi_2(s_2) &= \left[\varphi_1(s_1)p_{12} + \varphi_1(s_2)p_{22}\right] b_2(r_2) \\ &= \left[0.0524(0.15) + 3.7588(0.75)\right] 11.1075 \\ &= \left[0.0079 + 2.8191\right] 11.1075 \\ &= 31.4006.\end{aligned}$$

3. **Termination:**

$$p(r_{1:2} \mid \lambda) = \varphi_2(s_1) + \varphi_2(s_2) = 0.7790 + 31.4006 = 32.1796. \tag{4.10}$$

The marginal likelihood is mainly associated with the distressed terminal state, since $\varphi_2(s_2)$ is much larger than $\varphi_2(s_1)$. This result is coherent with the emission structure: two large negative returns are more likely under the high-volatility distressed regime.

Note: The forward recursion solves Problem 1 by computing the likelihood of the observed sequence[18]. For local decoding, it is combined with a backward recursion in the *Forward-Backward* algorithm to yield smoothed posterior probabilities, quantifying the probability of each hidden regime at time t given observed returns.

The Viterbi Algorithm

This algorithm[17] solves the decoding problem as it reconstructs the most probable sequence of latent regimes given the observed data. The main difference is that, while the Forward algorithm *sums* over all possible hidden-state paths, the Viterbi algorithm *maximizes* over them. At each time t and for each state j, it keeps only the most probable path ending in that state. To reconstruct the optimal path, a backpointer indicates from which state the maximum likelihood was obtained[18].

Definition 4.5 (The Viterbi Variable). Given an HMM parameterized by $\lambda = (P, B, \alpha)$, the Viterbi variable $\delta_t(i)$ is the maximum likelihood of observing the return sequence $r_{1:t} = (r_1, r_2, \ldots, r_t)$ following a path ending in state s_i at time t:

$$\delta_t(i) \triangleq \max_{x_{1:t-1}} p(x_{1:t-1}, X_t = s_i, r_{1:t} \mid \lambda). \tag{4.11}$$

where $x_{1:t-1}$ denotes a sequence of hidden states from time 1 to time $t-1$.

The most likely hidden-state sequence is obtained by the following steps:

1. **Initialization** ($t = 1$): for each hidden state $i \in \{1, \ldots, K\}$, the algorithm computes the joint probability of starting in state i and observing the first r_1:

$$\delta_1(i) = \alpha_i b_i(r_1), \qquad \psi_1(i) = 0. \tag{4.12}$$

   $\psi_1(i) = 0$ indicates that there is no previous state at the initial time step.

2. **Recursion** ($t = 2, 3, \ldots, T$): for each hidden state j, it selects the most probable path leading to state j at time t, maximizing over all possible states i:

$$\delta_t(j) = \left[\max_{1 \le i \le K} \delta_{t-1}(i) p_{ij}\right] b_j(r_t). \tag{4.13}$$

   The backpointer stores the previous state that achieved the maximum:

$$\psi_t(j) = \arg\max_{1 \le i \le K} \left[\delta_{t-1}(i) p_{ij}\right] \tag{4.14}$$

3. **Termination**: at the final time step T, the maximum joint likelihood of the optimal path is obtained by selecting the largest final Viterbi variable:

$$P^* = \max_{1 \le i \le K} \delta_T(i). \tag{4.15}$$

The final state of the optimal path is:

$$X_T^* = \arg\max_{1 \le i \le K} \delta_T(i). \tag{4.16}$$

4. **Path backtracking**: the full optimal sequence is reconstructed backwards:

$$X_t^* = \psi_{t+1}(X_{t+1}^*), \qquad t = T-1, T-2, \ldots, 1. \tag{4.17}$$

Application: Viterbi Algorithm with Gaussian Emissions

Consider the same HMM used in the previous example, with latent regimes $\mathcal{S} = \{s_1, s_2\}$, where $s_1 =$ Stable and $s_2 =$ Distressed and returns: $r_{1:2} = (-0.035, -0.028)$. The objective is to identify the most likely hidden-state path, denoted by $X_{1:2}^*$.

1. **Initialization** ($t = 1$):

$$\delta_1(s_1) = \alpha_1 b_1(r_1) = 0.6 \cdot 0.0873 = 0.0524,$$
$$\delta_1(s_2) = \alpha_2 b_2(r_1) = 0.4 \cdot 9.3971 = 3.7588.$$

$$\psi_1(s_1) = 0, \qquad \psi_1(s_2) = 0.$$

2. **Recursion** ($t = 2$):

For the *stable* regime:

$$\begin{aligned}
\delta_2(s_1) &= \max\left[\delta_1(s_1)p_{11}, \delta_1(s_2)p_{21}\right] b_1(r_2) \\
&= \max\left[0.0524(0.85), 3.7588(0.25)\right] 0.7915 \\
&= \max\left[0.0445, 0.9397\right] 0.7915 \\
&= 0.7438.
\end{aligned}$$

Since the maximum is achieved by the transition from s_2, the backpointer is:

$$\psi_2(s_1) = s_2.$$

For the *distressed* regime:

$$\begin{aligned}\delta_2(s_2) &= \max\left[\delta_1(s_1)p_{12}, \delta_1(s_2)p_{22}\right] b_2(r_2) \\ &= \max\left[0.0524(0.15), 3.7588(0.75)\right] 11.1075 \\ &= \max\left[0.0079, 2.8191\right] 11.1075 \\ &= 31.3133.\end{aligned}$$

Since the maximum is achieved by the transition from s_2, the backpointer is:

$$\psi_2(s_2) = s_2.$$

3. **Termination:**

$$P^* = \max[\delta_2(s_1), \delta_2(s_2)] = \max[0.7438,31.3133] = 31.3133.$$

$$X_2^* = s_2.$$

4. **Path Backtracking:** The previous state is reconstructed by the backpointer:

$$X_1^* = \psi_2(X_2^*) = \psi_2(s_2) = s_2.$$

Hence, the Viterbi algorithm identifies the most likely hidden-state sequence as:

$$X_{1:2}^* = (s_2, s_2).$$

The two consecutive negative returns are classified as more compatible with the *distressed* regime. This result is consistent with the Gaussian structure, where the distressed state has higher volatility and assigns larger density to negative returns.

4.4 The Baum-Welch Estimation Algorithm

The algorithms introduced in the previous section assume that the HMM parameters $\lambda = (P, B, \alpha)$ are known. However, in empirical applications, these parameters are not observable. In financial modeling, the transition probabilities between latent regimes and the associated emission densities must be estimated from data[18, 17].

The Baum-Welch algorithm addresses *Problem 3* by estimating the parameters of an HMM from an observed sequence. It is an iterative Expectation-Maximization procedure where the parameters are updated to increase the likelihood of the data.

1. **Initialization:** the algorithm starts from an initial parameter estimate:

$$\lambda^{(0)} = (P^{(0)}, B^{(0)}, \alpha^{(0)}).$$

2. **Expectation Step:** given the estimate $\lambda^{(r)}$, the algorithm computes the forward variables $\varphi_t^{(r)}(i)$, using the *Forward recursion* introduced in Section 4.3, the backward variables $\beta_t^{(r)}(i)$ and posterior quantities $\gamma_t^{(r)}(i)$ and $\xi_t^{(r)}(i,j)$[18].

Let us introduce the backward variable[18], accounting for future observations:

Definition 4.6 (The Backward Variable). Given an HMM parameterized by $\lambda^{(r)} = (P^{(r)}, B^{(r)}, \alpha^{(r)})$, the backward variable $\beta_t^{(r)}(i)$ is the likelihood of observing the future return sequence from time $t+1$ to time T, given the state at time t, s_i:

$$\beta_t^{(r)}(i) \triangleq p(r_{t+1:T} \mid X_t = s_i, \lambda^{(r)}). \tag{4.18}$$

The backward variable is computed recursively through two steps:

- **Initialization** ($t = T$): for each hidden state $i \in \{1, \ldots, K\}$, $\beta_T^{(r)}(i) = 1$.
- **Recursion** ($t = T-1, T-2, \ldots, 1$): for each state $i \in \{1, \ldots, K\}$, the algorithm sums over all possible next states j. Each term considers the probability of moving from state i to state j, the density of observing r_{t+1} from state s_j and the probability of explaining the future observations from state j:

$$\beta_t^{(r)}(i) = \sum_{j=1}^{K} p_{ij}^{(r)} b_j^{(r)}(r_{t+1}) \beta_{t+1}^{(r)}(j).$$

Once the forward and backward variables have been computed, they can be combined to obtain the posterior quantities $\gamma_t^{(r)}(i)$ and $\xi_t^{(r)}(i,j)$, which describe the expected state occupancy and expected state transition under the current parameter estimate.

Note: The posterior quantities computed through Forward-Backward have a double role. They provide local regime probabilities for inference and, inside Baum-Welch, they become expected state occupancies and transitions used to update parameters.

Definition 4.7 (State Occupancy Probability). The quantity $\gamma_t^{(r)}(i)$ is the posterior probability of being in hidden state s_i at time t, given the observed return sequence:

$$\gamma_t^{(r)}(i) \triangleq \mathbb{P}(X_t = s_i \mid r_{1:T}, \lambda^{(r)}). \tag{4.19}$$

It can be computed as:

$$\gamma_t^{(r)}(i) = \frac{\varphi_t^{(r)}(i)\beta_t^{(r)}(i)}{p(r_{1:T} \mid \lambda^{(r)})}. \tag{4.20}$$

The numerator combines the likelihood $\varphi_t^{(r)}(i)$ of explaining the observations up to time t and ending in state i with the likelihood $\beta_t^{(r)}(i)$ of the future observations from state i. The denominator normalizes it by the total likelihood of the sequence:

$$p(r_{1:T} \mid \lambda^{(r)}) = \sum_{i=1}^{K} \varphi_T^{(r)}(i). \tag{4.21}$$

Definition 4.8 (Expected Transition Probability). The quantity $\xi_t^{(r)}(i,j)$ is the posterior probability of being in state s_i at time t and moving to state s_j at time $t+1$, given the full observed return sequence and the current parameter estimate:

$$\xi_t^{(r)}(i,j) \triangleq \mathbb{P}(X_t = s_i, X_{t+1} = s_j \mid r_{1:T}, \lambda^{(r)}). \tag{4.22}$$

It can be computed as:

$$\xi_t^{(r)}(i,j) = \frac{\varphi_t^{(r)}(i)p_{ij}^{(r)}b_j^{(r)}(r_{t+1})\beta_{t+1}^{(r)}(j)}{p(r_{1:T} \mid \lambda^{(r)})}. \tag{4.23}$$

While $\gamma_t^{(r)}(i)$ measures the expected occupancy of state i at time t, $\xi_t^{(r)}(i,j)$ measures the expected transition from state i to state j between two consecutive time steps.

3. **Maximization Step**: the model parameters are then updated by the expected state occupancies and expected transitions computed in the previous step[18, 16].

$$\alpha_i^{(r+1)} = \gamma_1^{(r)}(i). \tag{4.24}$$

This means that the new initial probability of state i is set equal to the posterior probability of starting in that state. The updated transition probabilities are:

$$p_{ij}^{(r+1)} = \frac{\sum_{t=1}^{T-1} \xi_t^{(r)}(i,j)}{\sum_{t=1}^{T-1} \gamma_t^{(r)}(i)}. \tag{4.25}$$

For Gaussian emissions, the emission densities are updated through posterior-weighted sample moments. The updated mean of state s_i is:

$$\mu_i^{(r+1)} = \frac{\sum_{t=1}^{T} \gamma_t^{(r)}(i) r_t}{\sum_{t=1}^{T} \gamma_t^{(r)}(i)}. \tag{4.26}$$

The updated variance of state s_i is:

$$\left(\sigma_i^2\right)^{(r+1)} = \frac{\sum_{t=1}^{T} \gamma_t^{(r)}(i) \left(r_t - \mu_i^{(r+1)}\right)^2}{\sum_{t=1}^{T} \gamma_t^{(r)}(i)}. \tag{4.27}$$

4. **Convergence**: the *Expectation* and *Maximization* steps are repeated until the increase in log-likelihood becomes sufficiently small, converging to tolerance ε:

$$\left|\log p(r_{1:T} \mid \lambda^{(r+1)}) - \log p(r_{1:T} \mid \lambda^{(r)})\right| < \varepsilon. \tag{4.28}$$

Baum-Welch completes the HMM framework when the transition probabilities and emission densities are unknown. Given the observed return sequence, it iteratively updates the parameters until a stable solution is reached. It is not an exact global optimizer: it increases likelihood, but generally converges to a local maximum.

Remark on Regime Interpretation.

The numerical labels assigned to hidden states have no economic meaning. For example, state s_1 is not necessarily the low-volatility regime simply because it is listed first. Permuting state labels produces an equivalent likelihood, provided that the corresponding rows of P and the emission densities are permuted consistently. Therefore, regimes must be interpreted after estimation. In the Gaussian-emission framework, this interpretation is performed by ordering the states according to their estimated variance σ_i^2, identifying low, medium and high-volatility regimes.

This concludes the theoretical presentation of Markov Chains and Hidden Markov Models with Gaussian emissions. The following empirical part applies these procedures directly to S&P 500 returns, using Baum–Welch for parameter estimation, Forward–Backward for posterior probabilities and Viterbi for regime decoding.*

*The Forward, Viterbi and Baum-Welch algorithms follow Rabiner [18], Van Handel [16] and Zucchini, MacDonald and Langrock [17]. Their applications are supported by Bhar and Hamori [15]. Numerical superscripts refer to the complete Bibliography at the end of this work.

Section III

Empirical Application and Analysis of Results

Chapter 5

Model Estimation and Implementation in Python

5.1 Dataset Selection and Preparation

The theoretical framework developed in the previous chapters requires an empirical financial time series on which the Hidden Markov Model can be estimated. Since the objective is to *identify latent volatility regimes*, the application focuses on the **S&P 500 Index**, already used in Chapter 1 to illustrate time-varying volatility, volatility clustering and fat tails. This choice ensures continuity between the descriptive evidence previously introduced and the Python regime-switching model implemented.

The empirical goal is to move from *observable* market data to *latent unobservable* volatility regimes. The observed daily price series is converted into **logarithmic returns** and then used as the *primary input* for Gaussian HMM estimation:

$$S_t \longrightarrow r_t \longrightarrow \lambda = (P, B, \alpha).$$

The transformation from S_t to r_t recalls Definition 1.2 regarding logarithmic returns[2], while the parameter triplet $\lambda = (P, B, \alpha)$ follows Definition 4.3, which defines the HMM architecture[17, 18]. Here, P governs regime persistence, B contains the Gaussian emission densities, and α represents the initial state distribution.

The dataset consists of **daily observations** of the S&P 500 Index in the period 1980–2024, downloaded from Yahoo Finance[19]. The observed raw variable is the price, denoted by S_t, whereas the model is estimated on the corresponding **logarithmic return sequence** $\{r_t\}_{t=1}^{T}$. Modeling log-returns rather than prices is consistent with Definition 1.2[2]: while prices reflect a non-stationary cumulative evolution, logarithmic returns isolate the incremental variation between daily observations.

Implementation	**Python Object**	**Algorithmic Role**
Data Retrieval	`yf.download`	Downloads the historical S&P 500 price series used as the observable market data.
Log-Returns	`np.log`, `shift(1)`	Transforms prices S_t into percentage logarithmic returns $\tilde{r}_t$.
HMM Estimation	`GaussianHMM.fit`	Applies Baum-Welch to estimate α, P and the Gaussian emission parameters.
Likelihood	`model.score`	Evaluates the fitted log-likelihood through the Forward procedure.
Inference	`predict_proba`	Computes smoothed posterior probabilities through Forward-Backward.
State Decoding	`predict`	Reconstructs the most likely hidden-state sequence through Viterbi decoding.
Regime Ordering	`np.argsort(std_raw)`	Assigns Low, Medium and High Volatility labels after estimation.

Table 5.1: Computational workflow of the Gaussian HMM implementation.

Observations are ordered chronologically, since the HMM interprets returns as a sequence generated by an underlying Markov process rather than as independent data points. After computing logarithmic returns, any non-finite observations are removed, ensuring that the final input is a **valid ordered sequence** of daily returns.

The resulting dataset provides the observable layer of the empirical model. The **volatility regime** itself is not directly recorded in the data: it must be inferred from returns through the algorithms introduced in Chapter 4, namely Baum-Welch for estimation, Forward-Backward for posterior probabilities and Viterbi for the most likely regime path. The next section therefore defines the return transformation and the construction of the Gaussian HMM used to estimate hidden market regimes.

5.2 Calculation of Returns and Construction of the HMM Model

This section translates the dataset defined in Section 5.1 into the **computational structures** required by the Gaussian HMM. While the full Python code contains all the diagnostic routines, the focus here is on data preprocessing and model construction. Therefore, only the code fragments for data retrieval, return computation and model instantiation are reported, deferring all graphical results to Chapter 6.

The computational framework relies on `yfinance` to retrieve historical market data from Yahoo Finance[19, 21] and `hmmlearn` for the `GaussianHMM` class used to estimate the Hidden Markov Model[20], supported by `NumPy` and `pandas` numerical libraries.

```
import numpy as np
import pandas as pd
import yfinance as yf
import matplotlib.pyplot as plt

from hmmlearn.hmm import GaussianHMM
```

The empirical specification targets the S&P 500 Index (`^GSPC`) over the 1980–2024 window. As previously introduced, this index of 500 leading U.S. companies serves as a robust proxy for the broader U.S. equity market. The state space is configured with three latent components ($K = 3$) to structurally differentiate *low*, *medium*, and *high* volatility regimes. Lastly, an annualization factor of 252 is defined, representing the standard approximation of trading days per calendar year.

```
# Constant parameters setup
N_STATES = 3
TRADING_DAYS = 252

# S&P 500 data retrieval and preprocessing pipeline
data_raw = yf.download("^GSPC", start="1980-01-01", end="2025-01-01")

# Isolate closing prices, remove missing values and order chronologically
data = (data_raw[["Close"]].rename(columns={"Close": "Price"})
        .dropna().sort_index())
```

Given the preprocessed observable price sequence S_t defined in Section 5.1, the first computational step is the **logarithmic return** transformation. The sequence $\{r_t\}$ is computed using `shift(1)`, which shifts S_t down by one row to align S_t and S_{t-1}. To improve numerical readability and interpretability during the estimation of the Gaussian emission parameters, the raw returns are scaled into percentage terms:

$$\tilde{r}_t = 100 \cdot r_t = 100 \cdot \ln\left(\frac{S_t}{S_{t-1}}\right)$$

```
# Computing log-returns from current and previous closing prices
data["LogReturn"] = np.log(data["Price"] / data["Price"].shift(1))

# Scaling returns to percentage terms
data["LogReturnPct"] = 100 * data["LogReturn"]

# Removal of undefined values generated by the lagged division
data = data.replace([np.inf, -np.inf], np.nan).dropna()

# Reshaping the observable sequence into a 2D input matrix
returns = data["LogReturnPct"].values.reshape(-1, 1)
```

With the $T \times 1$ log-return matrix computed, the architecture is instantiated as a Gaussian HMM. Recalling Definition 4.2, each hidden state is associated with a state-dependent Gaussian emission density. Here, the latent process X_t takes values in the finite state space $\{s_1, s_2, s_3\}$, while the observed percentage return satisfies:

$$\tilde{r}_t \mid X_t = s_j \sim \mathcal{N}(\mu_j, \sigma_j^2), \qquad j = 1,2,3.$$

At this stage, the components s_j are purely abstract numerical indices. Their economic interpretation as *low*, *medium*, and *high* volatility regimes is assigned only after the estimation converges, by ordering the states according to their variance σ_j^2.

The unknown parameters, as formalized within the *parameter calibration problem* of Section 4.2, are the initial distribution α, the transition matrix P, and the emission parameters B. Therefore, the **Baum-Welch algorithm** estimates the transition probabilities, the initial probabilities and the state-specific means and variances.

The objective is to estimate the parameters maximizing the likelihood of the observed log-returns. As discussed in Section 4.4, Baum-Welch performs this iteratively, though its final solution is sensitive to initialization parameters[17, 18]. To mitigate this, the implementation employs a multi-restart fitting strategy across multiple random seeds, ultimately retaining the candidate that maximizes the log-likelihood.

```
def fit_gaussian_hmm_with_restarts(
    X,
    n_states,
    random_states=range(10),
    covariance_type="diag",
    n_iter=1000,
    tol=1e-4
):
    # Storage for the best fitted model
    best_model = None
    best_log_likelihood = -np.inf
    restart_summary = []

    for seed in random_states:
        # Same HMM specification, different initialization
        model = GaussianHMM(
            n_components=n_states,
            covariance_type=covariance_type,
            n_iter=n_iter,
            tol=tol,
            random_state=seed
        )

        # Estimate alpha, P, means and variances through Baum-Welch
        model.fit(X)
        log_likelihood = model.score(X)

        # Save convergence information for this restart
        restart_summary.append([
            seed,
            log_likelihood,
            model.monitor_.converged,
            model.monitor_.iter
        ])
```

```
        # Keep the model with the highest log-likelihood
        if log_likelihood > best_log_likelihood:
            best_log_likelihood = log_likelihood
            best_model = model

    restart_summary = pd.DataFrame(
        restart_summary,
        columns=["seed", "log_likelihood", "converged", "iterations"]
    )

    return best_model, restart_summary
```

The input X is the $T \times 1$ matrix constructed above. Inside the routine, `fit` applies Baum-Welch to estimate α, P and the Gaussian emission parameters, while `score` evaluates the log-likelihood of the observed sequence under the fitted candidate. The variables `converged` and `iterations` are stored only as numerical diagnostics, in order to check whether the iterative procedure has reached its stopping criterion.

Specifically, `covariance_type="diag"` matches the univariate return input by estimating a distinct variance σ_j^2 for each state. The stopping behavior of Baum-Welch is controlled by `n_iter` and `tol`, while `random_state` changes the initialization.

At the end of this section, the computational structure required for the empirical analysis is fully complete. The S&P 500 closing prices have been efficiently transformed into a one-column matrix of percentage log-returns, the three-state Gaussian HMM has been specified and a restart-based estimation routine has been defined.

The next section executes this pipeline to extract the core outputs: the fitted log-likelihood, the initial distribution α, the state-specific Gaussian parameters (μ_j, σ_j^2), the transition matrix P, the smoothed posterior probabilities obtained through Forward-Backward and the optimal hidden-state sequence via Viterbi decoding.

5.3 Model Estimation and Output Construction

Following the setup of Section 5.2, this section presents the core executable code used to estimate the model, extract the fitted parameters, decode hidden regimes and prepare the core empirical outputs for Chapter 6. The focus remains strictly computational, deferring all economic interpretation to the subsequent chapter.

The estimation *is triggered* by calling the multi-restart routine on the $T \times 1$ input matrix `returns` with $K = 3$ hidden states. Retaining the candidate maximizing the log-likelihood, this call outputs the selected fitted instance as `model`, alongside `restart_results`, recording the convergence status of every initialization attempt.

```
# Final estimation of the three-state Gaussian HMM
model, restart_results = fit_gaussian_hmm_with_restarts(
    returns,
    n_states=N_STATES,
    random_states=range(20),
    covariance_type="diag",
    n_iter=1000,
    tol=1e-4
)

# Extract convergence metrics for the optimal candidate
log_likelihood = model.score(returns)
converged = model.monitor_.converged
iterations = model.monitor_.iter

# Rank all initialization attempts to assess stability
restart_results = restart_results.sort_values(
    "log_likelihood",
    ascending=False
)
```

Once fitted, the calibrated parameters (α, P, B) are retrieved directly from the attributes of the `model` instance. Specifically, `startprob_raw` (α) and `transmat_raw` (P) store the *initial and transition probabilities*, while `means_raw` and `variances_raw` capture the *emission parameters* (μ_j, σ_j^2). The suffix `_raw` highlights that these components still represent arbitrary numerical indices, prior to economic sorting.

```
# Extract raw estimated HMM parameters
startprob_raw = model.startprob_
transmat_raw = model.transmat_
means_raw = model.means_.flatten()

# Extract state-specific variances and standard deviations
covars_raw = np.array(model.covars_).reshape(N_STATES, -1)
variances_raw = covars_raw[:, 0]
std_raw = np.sqrt(variances_raw)
```

Having estimated the theoretical parameters, the analysis shifts to **hidden state reconstruction**. Regime inference addresses the decoding problems formalized in Chapter 4[17, 18], yielding two fundamentally complementary outputs. The `predict` method solves the *global decoding* problem via the Viterbi algorithm, assigning the single most likely regime per trading day. Conversely, `predict_proba` addresses the *local decoding* problem via the Forward-Backward algorithm, computing the smoothed posterior probability of belonging to each latent regime at every step.

```
# Global decoding (Viterbi): optimal deterministic hidden-state path
hidden_states_raw = model.predict(returns)

# Local decoding (Forward-Backward): smoothed posterior probabilities
posterior_probs_raw = model.predict_proba(returns)

# Append the deterministic Viterbi classification to the dataset
data_model = data.copy()
data_model["RawState"] = hidden_states_raw
```

These two objects serve distinct empirical purposes. The Viterbi sequence provides a deterministic classification, strictly assigning each daily return to a specific state. Conversely, posterior probabilities quantify classification uncertainty by distributing probability mass across the K latent regimes. Together, they establish which volatility state prevails on a given date and how uncertain that membership is.

The 1D Viterbi vector is immediately appended to the dataset as a single column to facilitate the upcoming state mapping. Meanwhile, the $T \times K$ posterior probability matrix is retained in memory to be processed later for probabilistic visualizations.

As discussed in Chapter 4, the numerical labels assigned by an HMM have no economic meaning by themselves. Therefore, to identify meaningful volatility regimes, the estimated states must be sorted by their standard deviation using `np.argsort(std_raw)`. The resulting ascending indices are then mapped through the `raw_to_ordered` to assign the *Low*, *Medium* and *High Volatility* labels.

```
# Order hidden states by estimated standard deviation
state_order = np.argsort(std_raw)

ordered_names = {
    0: "Low Volatility",
    1: "Medium Volatility",
    2: "High Volatility"
}

# Map raw arbitrary indices to economically ordered indices
raw_to_ordered = {
    raw_state: ordered_state
    for ordered_state, raw_state in enumerate(state_order)
}

# Add ordered state labels to the empirical dataset
data_model["State"] = data_model["RawState"].map(raw_to_ordered)
data_model["Regime"] = data_model["State"].map(ordered_names)
```

Once the state ordering is established, all estimated parameters must be rearranged to reflect this hierarchy. This realignment is strictly an ex-post operation: it requires first fitting the model to the data to compute the standard deviations ($\hat{\sigma}_j$), which then dictate the consistent reordering of all outputs for the empirical tables.

```
# Reorder initial probabilities
startprob_ordered = startprob_raw[state_order]

# Permute the K x K transition matrix
transmat_ordered = transmat_raw[np.ix_(state_order, state_order)]

# Reorder 1D Gaussian emission parameters
means_ordered = means_raw[state_order]
variances_ordered = variances_raw[state_order]
```

```
std_ordered = std_raw[state_order]

# Reorder the state columns of the T x K posterior probability matrix
posterior_probs_ordered = posterior_probs_raw[:, state_order]
```

With parameters properly ordered, the numerical outputs are collected into DataFrames. This step structures the state-specific moments, transition dynamics and regime durations into tables that will be reported and interpreted in Chapter 6.

```
# State-specific Gaussian emission parameters
parameter_table = pd.DataFrame({
    "Regime": [ordered_names[i] for i in range(N_STATES)],
    "Mean (%)": means_ordered,
    "Std. Dev. (%)": std_ordered,
    "Variance": variances_ordered,
    "Annualized Volatility (%)": std_ordered * np.sqrt(TRADING_DAYS),
    "Initial Probability": startprob_ordered
})

# Transition matrix ordered by volatility regime
transition_table = pd.DataFrame(
    transmat_ordered,
    index=[ordered_names[i] for i in range(N_STATES)],
    columns=[ordered_names[i] for i in range(N_STATES)]
)

# Expected duration implied by the self-transition probabilities
expected_duration = pd.DataFrame({
    "Regime": [ordered_names[i] for i in range(N_STATES)],
    "Self-transition Probability": np.diag(transmat_ordered),
    "Expected Duration (days)": 1 / (1 - np.diag(transmat_ordered))
})
```

The expected duration represents the average number of consecutive trading days spent in the same regime before switching to another state. If p_{jj} denotes the daily self-transition probability of state s_j, then the expected duration is given by[17]:

$$\mathbb{E}(D_j) = \frac{1}{1 - p_{jj}}.$$

This quantity thus provides a highly useful empirical measure of **regime persistence** and will be used in Chapter 6 to discuss the evidence of volatility clustering.

Since the Viterbi algorithm assigns *one most likely hidden state* to each trading day, the **relative frequency** provides a clear and compact summary of how often each volatility regime appears across the full observation period considered.

```
# Empirical frequency of each Viterbi regime
regime_frequency = (
    data_model["Regime"]
    .value_counts(normalize=True)
    .reindex([ordered_names[i] for i in range(N_STATES)])
    .rename("Frequency")
    .to_frame()
)
```

The resulting DataFrames organize the estimated quantities for subsequent analysis. The Gaussian parameters describe the return distribution for each hidden state, the transition matrix and expected durations summarize regime persistence, and the Viterbi frequencies show how often each regime appears in the decoded path.

The Python implementation pipeline is therefore complete: the S&P 500 three-state Gaussian HMM has been estimated, decoded and organized into final outputs that will be reported and interpreted in Chapter 6 through tables and figures.*

*The empirical data source for the S&P 500 prices is Yahoo Finance [19], accessed through `yfinance` [21]. The HMM implementation relies on `hmmlearn` [20]. Likelihood evaluation, posterior inference, Viterbi decoding and Baum-Welch estimation recall Rabiner [18] and Zucchini, MacDonald and Langrock [17]. Numerical superscripts refer to the complete Bibliography.

Chapter 6

Regime Interpretation and Empirical Evaluation

6.1 Identification of Volatility Regimes

The Python implementation in Chapter 5 produces the estimated parameters of the three-state Gaussian HMM. The objective is to examine whether the latent structure of S&P 500 returns corresponds to **economically interpretable** volatility regimes.

This interpretation closes the link developed throughout the thesis. The time-varying behavior observed in Chapter 1 was connected in Chapter 2 to the presence of unobservable market conditions, and later formalized in Chapter 4 through a Gaussian HMM with state-dependent return distributions. For this reason, the first empirical evidence to examine is the set of estimated moments associated with each hidden state: *mean, standard deviation, variance and annualized volatility.* The yearly risk measure is computed by rescaling the standard deviation of each regime for the approximate number of trading days in one year: $\hat{\sigma}_{ann,j} = \hat{\sigma}_j\sqrt{252}$.

Regime	**Mean (%)**	**Std. Dev. (%)**	**Variance**	**Annualized Vol. (%)**
Low Volatility	0.089	0.593	0.351	9.410
Medium Volatility	0.007	1.140	1.301	18.104
High Volatility	-0.213	2.956	8.736	46.919

Table 6.1: Estimated Gaussian emission parameters ordered by volatility regime.

Table 6.1 translates the hidden states into measurable financial quantities. Recalling Definition 1.4[2], the mean $\hat{\mu}$ summarizes the central tendency of returns, while the variance $\hat{\sigma}^2$ measures their dispersion around that level. In this section, the standard deviation $\hat{\sigma}_j$ is the key volatility indicator used to compare the hidden regimes.

The *Low Volatility* regime is associated with the smallest dispersion, with a standard deviation of 0.593%, and a slightly positive mean return of 0.089%. It represents the *calmest market environment*, where daily fluctuations remain quite contained.

The *Medium Volatility* regime occupies an intermediate position. Its average return is close to zero, equal to 0.007%, while its daily standard deviation rises to 1.140%, almost twice that of the low-volatility state. This suggests a *normal but less stable* market condition, where uncertainty is higher without implying extreme distress.

The *High Volatility* regime is separated from the other two states. Its standard deviation reaches 2.956% and its estimated mean is negative, equal to -0.213%, suggesting that *the most turbulent regime* is associated with adverse market movements.

The separation between the estimated regimes is consistent with the main argument developed in Chapter 1: a single unconditional variance cannot adequately summarize the full return series, as it compresses calm and turbulent conditions into one aggregate distribution. The HMM addresses this limitation by decomposing returns into state-dependent Gaussian components, each with its own level of dispersion.

The variance difference is particularly informative: moving from the low to the high-volatility regime, the estimated variance increases from 0.351 to 8.736. This nearly twenty-five-fold jump shows a clear separation between the volatility regimes. While mean estimates require caution, as the model primarily separates states by variance, the negative mean of the high-volatility state broadly aligns with the financial intuition that market stress typically coincides with adverse price movements.

The first conclusion is that the Gaussian HMM represents the apparently single S&P 500 return process into *three interpretable latent volatility regimes*. This result provides the basis for the next step of the analysis: verifying whether these regimes display temporal persistence, providing empirical support for volatility clustering.

6.2 Regime Persistence and Classification Uncertainty

After identifying the volatility regimes via their Gaussian parameters, the next step is analyzing their *temporal persistence*. The model examines whether these states persist over time, consistently with the *volatility clustering* discussed in Chapter 1: large fluctuations follow large fluctuations, while calm periods remain stable.

The first object to examine is the estimated **transition matrix**. Recalling Chapter 3, each element p_{ij} represents the probability of moving from state i to state j in one step. Since one step corresponds to one trading day, the diagonal elements p_{jj} measure the probability of remaining in the same volatility regime the next day.

	Low Volatility	**Medium Volatility**	**High Volatility**
Low Volatility	0.979	0.021	0.000
Medium Volatility	0.025	0.969	0.006
High Volatility	0.000	0.044	0.956

Table 6.2: Estimated transition matrix ordered by volatility regime.

Table 6.2 shows a strong concentration of probability mass along the diagonal, suggesting that volatility conditions are locally persistent: once the market enters a specific state, the model assigns a high probability to remaining there the next day.

The diagonal information can also be directly translated into expected regime duration, as defined in Chapter 5. This measure expresses intuitively how long each underlying volatility condition tends to last in terms of consecutive trading days.

Regime	**Self-transition**	**Expected Duration**	**Viterbi Frequency**
Low Volatility	0.979	47.33 days	52.03%
Medium Volatility	0.969	32.50 days	42.59%
High Volatility	0.956	22.48 days	5.39%

Table 6.3: Persistence and frequency of the estimated volatility regimes.

Table 6.3 combines two complementary measures: expected duration and Viterbi frequency, the percentage of the sample spent in each regime. The *low-volatility* regime is the most persistent and frequent, lasting on average 47.33 days and covering 52.03% of the observations. The *medium-volatility* regime also plays a central role, with an expected duration of 32.50 days and a frequency of 42.59%.

The *high-volatility* regime is rare, appearing in only 5.39% of the Viterbi path, but not economically marginal. Its expected duration of 22.48 days indicates that turbulent phases are not isolated daily shocks: once reached, they tend to last for several consecutive sessions, consistent with the volatility clustering phenomenon.

The transition matrix shows that direct switches between low and high volatility are almost absent. Transitions mainly occur through the medium-volatility regime, which acts as an intermediate layer between calm and stressed market conditions.

The plot shown in Figure 6.1 reports the Viterbi classification of daily log-returns. Each point represents one observed trading day, while its color indicates the regime assigned by the most likely hidden-state sequence. The classification is therefore deterministic: although the true regime is latent, the Viterbi algorithm reconstructs *one most likely path* and associates each observation with a single volatility state.

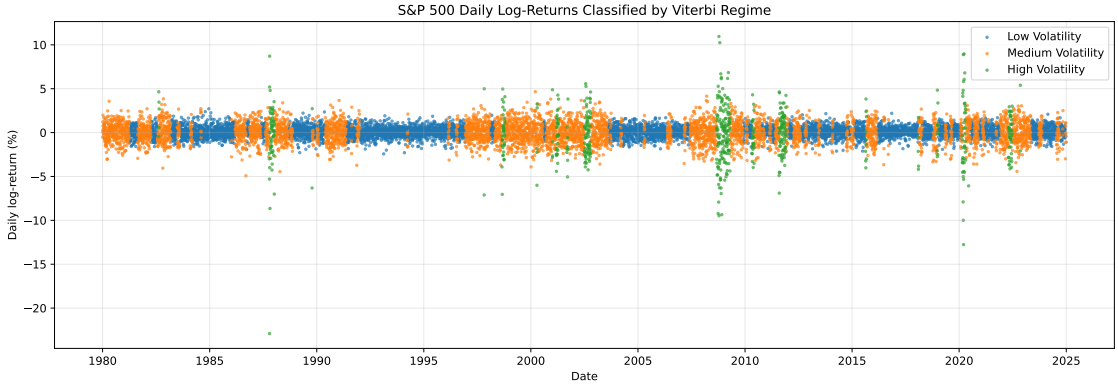


Figure 6.1: *S&P 500 daily log-returns classified by Viterbi volatility regime.*

The visual pattern is consistent with parameters estimated in Section 6.1. Returns classified as *low volatility* remain concentrated around small oscillations, while *high-volatility* observations correspond to larger movements, indicating instability.

Similar colors cluster in consecutive intervals, suggesting that the volatility states are temporally persistent. This aligns with the transition matrix, where switches between low and high are almost absent, crossing the medium-volatility regime.

The Viterbi representation hides assignment uncertainty because every trading day receives a single label, making historical reconstruction rigid. Posterior probabilities therefore show that the model does not remove uncertainty, but represents it probabilistically through Forward-Backward inference, introduced in Chapter 4.

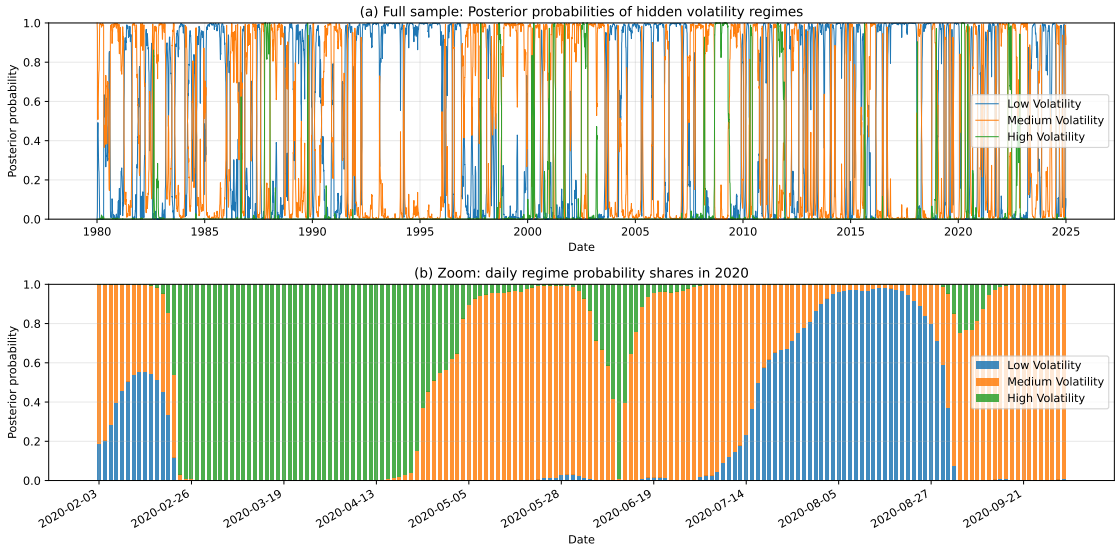


Figure 6.2: *Posterior probabilities of the hidden volatility regimes. (a) The full sample shows the evolution of regime probabilities over the time window. (b) The zoomed 2020 market-stress period represents each trading day as a stacked bar.*

The plot illustrates the evolution of posterior probabilities over time. Each curve corresponds to one hidden regime and, for each day, the three probabilities sum to one. The zoomed specifically during the volatile 2020 market-stress period uses stacked bars to detail how probability mass is distributed. When one regime dominates, the model classifies the observation with high confidence; when probability mass splits, classification becomes uncertain, often around regime changes.

This representation complements the Viterbi path. The regimes are *differentiated*, *persistent* and *probabilistically interpretable*. Their persistence is consistent with volatility clustering, motivating the comparison with a Gaussian benchmark.

6.3 Comparison with a Static Gaussian Model

After having identified regimes and analyzed their temporal persistence, the final analytical step evaluates the *distributional effect* of the HMM, assessing how well the estimated model approximates real returns. The objective is to compare the empirical distribution of S&P 500 daily log-returns with two alternatives: a **static Gaussian density** and the **Gaussian mixture** implied by the estimated HMM.

This comparison recalls the limitation discussed in Chapter 1, that a static Gaussian model summarizes the entire return series through one unconditional mean and variance. As a consequence, calm periods and turbulent periods are compressed into a single distribution, thereby ignoring the distinction between volatility regimes.

Conversely, from Definition 1.6[5], the HMM addresses this compression by constructing a sample-based mixture of the fitted Gaussian components. These correspond to the low, medium, and high-volatility regimes from Section 6.1. Using the sample averages of the smoothed posterior probabilities as mixture weights, this mixture reflects the contribution of each estimated regime to the observed return distribution.

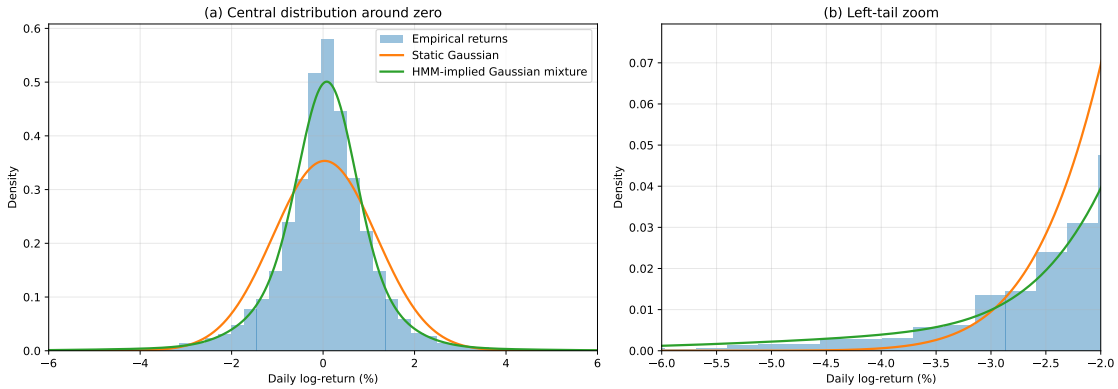


Figure 6.3: *S&P 500 daily log-returns compared with a static Gaussian and the three-state Gaussian mixture implied by the HMM. (a) Full view shows the distribution around zero. (b) Zoomed view focuses on the left tail to highlight extreme returns.*

This histogram compares the empirical distribution of daily log-returns with two model-based approximations. The *static Gaussian* benchmark provides a single symmetric density centered around the mean. Its limitation is visible in the central part of the distribution: the curve appears too flat relative to the concentration of returns around zero, while failing to adequately represent extreme observations.

The HMM-implied mixture provides a more flexible approximation because it combines the three state-dependent Gaussian distributions estimated by the model. The low-volatility component contributes to the central part of the mixture around zero returns, whereas the high-volatility component assigns more probability mass to large deviations. The resulting curve therefore reflects *more effectively* the coexistence of frequent small movements and rare but substantial market shocks.

As expected, the approximation does not fit details of the empirical histogram. The HMM does not reproduce the central peak *perfectly* because it remains a statistical model based on a finite number of Gaussian regimes. Its purpose is not to mirror every irregularity of observed data, but to provide a structured description of the main sources of market uncertainty. In this sense, the model moves closer to the empirical behavior of returns without claiming to capture reality in its entirety.

This result clarifies the true value of the *regime-switching* framework: the presence of fat tails does not require abandoning Gaussian distributions entirely. By maintaining local Gaussianity within each regime, the aggregation of different variances naturally produces an overall distribution closer to the sharper central peak and heavier tails of real returns. The crucial improvement of the HMM is that these local components are not treated as static, independent groups; rather, they are *dynamically linked* by the latent Markovian transitions previously analyzed in Section 6.2.

Consequently, the HMM provides a more flexible representation of S&P 500 returns than a single rigid density. It retains the interpretability of Gaussian emissions while allowing the global distribution to reflect distinct, persistent volatility phases. This concludes the empirical evaluation, paving the way for the final section to summarize how the model jointly addresses the core stylized facts of the series.

6.4 Empirical Verification and Final Conclusions

The final step returns to the stylized facts introduced in Chapter 1. After identifying volatility regimes, analyzing their persistence and comparing the HMM-implied distribution with a static Gaussian benchmark, the remaining question is whether the estimated model is coherent with the original evidence of volatility clustering.

For this purpose, the final diagnostic step recalls Definition 1.5[1], where the sample **autocorrelation function** was introduced to measure linear dependence between a time series and its lagged values. Applied to raw log-returns, the ACF assesses whether the direction of returns displays material linear serial dependence. As discussed in Chapter 1, this dependence is expected to remain *close to zero*, meaning past returns provide limited information about future market direction.

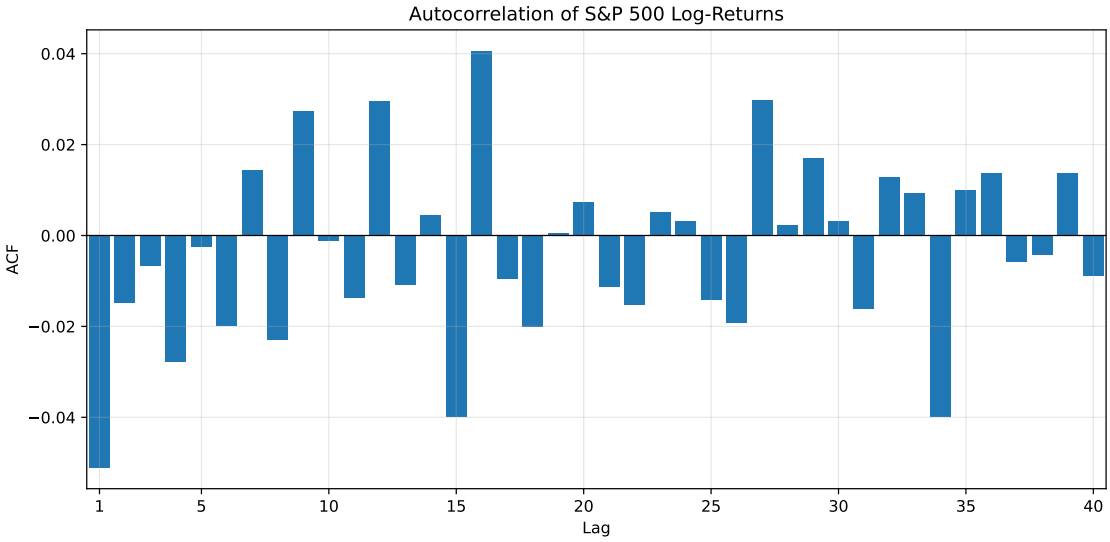


Figure 6.4: *Autocorrelation function of S&P 500 daily log-returns.*

Lag zero is omitted because its autocorrelation equals one by construction; the displayed lags therefore focus on the meaningful dependence structure. This is consistent with the martingale intuition discussed in Chapter 2[9, 10]: past returns provide limited information about future direction. Persistence must therefore be searched through volatility, which concerns magnitude rather than sign: a positive spike does not imply another positive return, but may signal moves of either sign.

The same diagnostic becomes more revealing when applied to **absolute log-returns**. By focusing on the *magnitude* rather than the direction of movements, a positive and slowly decaying autocorrelation indicates that days with similar volatility tend to cluster together. Therefore, comparing these ACFs separates directional unpredictability from volatility persistence, demonstrating the clustering effect.

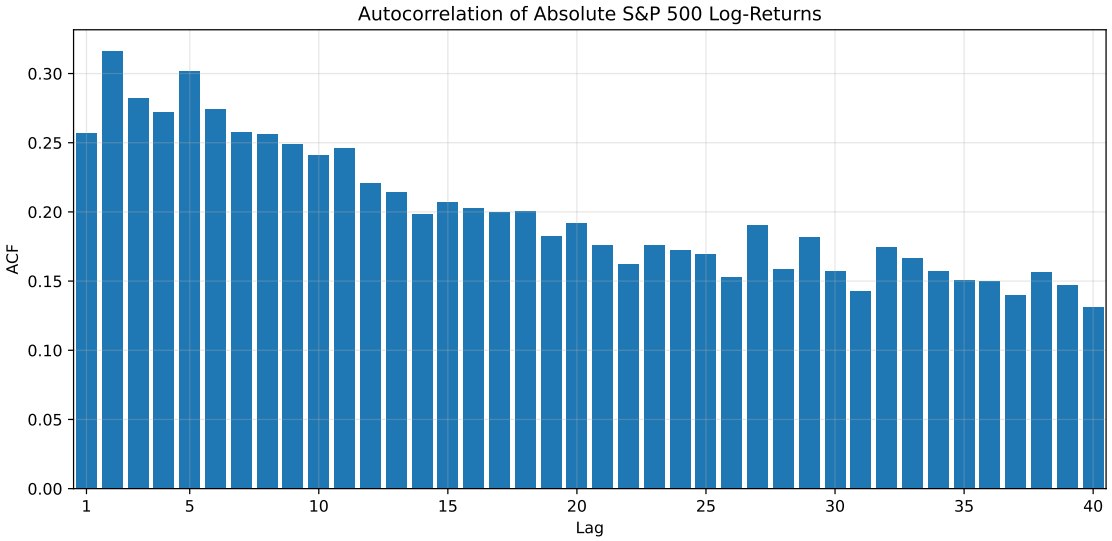


Figure 6.5: *Autocorrelation function of S&P 500 absolute daily log-returns.*

The informative lags of the absolute-return ACF confirm this interpretation: unlike the raw-return case, they remain positive over several days instead of collapsing immediately around zero. This persistence provides the *empirical* counterpart of the regime durations estimated in Section 6.2, where volatility states were shown to last for several consecutive trading days in the observed data, as a measurable signal.

Taken together, the two ACF diagnostics show that persistence is clearly stronger in return magnitude than in return sign. This pattern is consistent with, rather than a direct validation of, the latent regime persistence estimated by the HMM.*

*The interpretation of autocorrelation, finite Gaussian mixtures and volatility clustering also recalls Mastrantonio [1], Oosterlee and Grzelak [5], Mandelbrot [4] and Cont [7]. The HMM interpretation follows Rabiner [18] and Zucchini, MacDonald and Langrock [17]. Numerical superscripts refer to the complete Bibliography at the end of this work.

Final Conclusions:

The overall analysis answers the thesis objective through three connected findings. First, the Gaussian HMM identifies *distinct volatility regimes*, characterized by different levels of variance. Second, these regimes are persistent, giving a probabilistic interpretation of *volatility clustering*. Third, their combination produces an *unconditional distribution* closer to empirical returns than a static Gaussian.

Thus, financial uncertainty cannot be represented by a single distribution with fixed parameters. The HMM instead represents time-varying return dynamics through measurable latent regimes. It does not eliminate uncertainty nor recover true market states with certainty; it *quantifies* this uncertainty through regime probabilities, transition dynamics and state-dependent volatility estimates.

The analysis is retrospective by design: past regimes are estimated to show that the model gives a coherent description of market uncertainty. In quantitative finance, however, history is the sample from which a probabilistic rule is learned. In a live setting, using data up to day t, filtered regime probabilities could be projected through the transition matrix to support risk management, sizing and allocation. A related application appears at the *European Stability Mechanism* (ESM)*, where in 2025 a Senior Portfolio Manager applied a Hidden Markov Model to US, German and French government bond data to build a portfolio optimisation strategy.

As with any statistical framework, the model must be interpreted as an *approximation* rather than a representation of market reality. The Gaussian assumption remains local to each regime, the number of states is fixed a priori and estimation depends on initialization. Therefore, the scope of this framework is to describe the core structure of regime-dependent uncertainty rather than eliminate market randomness.

The thesis concludes that **Hidden Markov Models** offer a rigorous, interpretable statistical approach for describing financial uncertainty. By representing time-varying return dynamics through latent regimes, the model connects stylized facts, empirical evidence and practical relevance within a unified probabilistic framework.

*Practical application based on Byran Taljaard, *Regime-based Portfolio Optimisation: A Hidden Markov Model Approach for Fixed Income Portfolios*, European Stability Mechanism, 2025.

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